

Ph.D. Thesis Proposal

**Spatial Perception that Generalized: Learning
Uncertainty from Vision to IMU**

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Abstract

The ultimate vision of autonomous robotics is to create systems that can adapt, learn, and operate robustly across diverse environments without constant human intervention. However, current SLAM (Simultaneous Localization and Mapping) systems remain brittle, requiring extensive manual tuning and failing when faced with novel scenarios or sensor configurations. This thesis addresses a fundamental question: "Can we create SLAM systems that automatically adapt to different platforms and sensors, ultimately making SLAM engineering obsolete?"

This dissertation presents a comprehensive framework for learning-based SLAM that advances toward fully automated spatial AI systems. The core contribution lies in developing data-driven approaches that replace traditional geometric methods with learned representations, enabling systems to generalize across platforms and automatically handle uncertainty.

The first part presents my completed work on learning-based odometry and uncertainty modeling. MAC-VO introduces a metrics-aware covariance approach for stereo visual odometry that won Best Paper at ICRA 2025, demonstrating how learned uncertainty estimates significantly improve system robustness. AirIO advances inertial odometry through enhanced IMU feature observability, while AirDOS explores dynamic SLAM that benefits from understanding articulated objects in the environment.

The second part outlines proposed research directions that push toward fully automated SLAM systems. This includes developing foundation models for spatial understanding that can automatically adapt to new sensor configurations, environments, and platforms without requiring manual parameter tuning or extensive dataset collection.

The proposed framework is evaluated on challenging benchmarks including the ICCV 2023 SLAM Challenge, SubT-MRS, LARMA, TUM, and EuRoC datasets. A novel robustness metric is introduced to assess system performance under sensor failure and environmental degradation.

Together, these efforts form a unified framework for building resilient, intelligent, efficient SLAM systems—designed to work for any robot, anywhere.

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Chapter 1

Introduction

Robotics faces a fundamental challenge: while we have sophisticated sensors and algorithms, current perception systems fail to generalize across diverse environments and platforms due to their reliance on hand-tuned uncertainty models. Traditional robotic perception approaches use fixed covariance matrices and empirically-set noise parameters that work well in controlled scenarios but break down when deployed in new environments, on different platforms, or under varying operational conditions.

This brittleness stems from a deeper problem: the inability to accurately model and propagate uncertainty through the perception pipeline. Current systems treat uncertainty as an afterthought, using simplistic models that fail to capture the true reliability of sensor measurements and their correlations across modalities.

My research addresses this fundamental limitation through a comprehensive framework for learning-based uncertainty estimation. Rather than relying on hand-crafted parameters, this dissertation demonstrates how data-driven uncertainty models can be learned from experience, adapted across platforms, and propagated through complex perception pipelines to enable truly generalizable robotic perception systems.

1.1 Motivation

1.2 Research Questions

This dissertation organizes the research questions across four parts:

1.2.1 Part I: Metrics-Aware Uncertainty for Visual Odometry

RQ1: How can we learn scale-consistent, metrics-aware visual uncertainty and leverage it to select and weight correspondences for robust VO/SLAM across cameras and environments?

1.2.2 Part II: Uncertainty-Aware Multi-Modal Fusion and Cross-Platform Adaptation

RQ2: How can we learn and calibrate a data-driven model of IMU preintegration uncertainty for pose-graph optimization?

RQ3: What IMU feature representations enable robust learning-based inertial odometry under high-dynamics (e.g., drones), and why do current approaches fail?

1.2.3 Part III: Uncertainty-Aware SLAM

RQ4: How can we leverage metrics-aware uncertainties from both vision and inertia to enable robust and versatile VIO without manual tuning?

RQ5: How can we design a robust SLAM system that explicitly models multi-frame measurement covariance in the factor graph to improve feature selection, loop closure, and estimator consistency across environments and platforms?

1.2.4 Part IV: Uncertainty-Driven Adaptation, Self-Supervision, and Representation Learning

RQ6: How can we develop self-supervised domain transfer methods for uncertainty models that enable zero-shot deployment with automatic sensor configuration?

RQ7: How can we learn contact states and their uncertainty via a differentiable EKF to improve legged IMU odometry?

RQ8: What IMU tokenization best conditions visual foundation models and inertial odometry, and how should uncertainty be incorporated into the tokens?

1.3 Part I: Metrics-Aware Uncertainty for Visual Odometry

1.4 Thesis Statement

Chapter	Contribution	Research Question	Status (Publications)
Chapter 3	C1	RQ1	Completed
Chapter 4	C2	RQ2	Completed
Chapter 5	C3	RQ3	Completed
Chapter 7	C4	RQ4	Proposed
Chapter 8	C6	RQ6	Proposed
Chapter 9	C7	RQ7	Proposed
Chapter 10	C8	RQ8	Proposed

Table 1.1: Contributions and research questions addressed in each chapter

Thesis Statement:

This thesis delivers a unified, uncertainty-driven spatial perception stack that converts visual and inertial measurements into *metrics-aware, calibrated beliefs*, enabling reliable correspondence, principled fusion, and consistent multi-frame SLAM across platforms and conditions (day/night, indoor/outdoor, high dynamics).

This work reframes uncertainty from a fixed, hand-tuned quantity to a learnable, metrics-aware signal that drives correspondence, factor weighting, and multi-frame estimation. Unlike approaches that require per-platform tuning, the framework learns and calibrates uncertainty from data and adapts it across sensors and environments—yielding a deployable spatial perception system that remains accurate under adverse illumination, motion, and domain shift, and interfaces cleanly with VFM and RL via learned inertial representations.

1.5 Thesis Contributions

This dissertation makes the following key contributions across four parts:

1.5.1 Part I: Metrics-Aware Uncertainty for Visual Odometry

C1. MAC-VO: Visual Perception Uncertainty Learning

[Completed]

MAC-VO introduces a metrics-aware covariance learning framework for stereo visual odometry that predicts uncertainty directly from visual features rather than relying on fixed geometric models. Trained across diverse cameras and lighting, MAC-VO yields scale-consistent uncertainty that reflects correspondence reliability and improves localization and fusion.

1.5.2 Part II: Uncertainty-Aware Multi-Modal Fusion and Cross-Platform Adaptation

C2. AirIMU: Inertial Uncertainty Learning

[Completed]

AirIMU develops a data-driven framework for learning IMU preintegration uncertainty that captures platform- and motion-specific noise characteristics without manual tuning. Using differentiable integration and covariance propagation, AirIMU enables accurate uncertainty modeling across robotic platforms.

C3. AirIO: Uncertainty-Aware Inertial Odometry

[Completed]

AirIO enhances inertial odometry by leveraging learned IMU uncertainty and improved observability, enabling robust operation in GPS-denied environments and reduced drift over long trajectories.

1.5.3 Part III: Uncertainty-Aware SLAM

C4. MACVIO: Cross-Modal Uncertainty Fusion

[Proposed]

{Template: Describe how MACVIO learns cross-modal uncertainty correlations to enable adaptive fusion weights, sensor selection, and improved consistency}

C5. MACSLAM Uncertainty-Aware SLAM

[Proposed]

{Template: Describe how the SLAM system models measurement covariance, cross-factor correlations, and calibration drift to improve robust data association, loop closure, and optimization}

1.5.4 Part IV: Uncertainty-Driven Adaptation, Self-Supervision, and Representation Learning

C6. MAC-IO: Uncertainty-Guided Self-Supervised Inertial Odometry

[Proposed]

{Template: Describe how predictive uncertainty calibrates pseudo-labels and objectives for stable self-supervised IO across platforms}

C7. Contact-Aware Legged IMU Odometry via Differentiable EKF

[Proposed]

{Template: Describe jointly learned contact states and uncertainty, and differentiable EKF improving observability and reducing drift}

C8. IMU Tokenizer for Vision Foundation Models and RL

[Proposed]

{Template: Describe transformer-based IMU tokenization aligned with vision tokens; uncertainty-calibrated embeddings for fusion and RL}

1.6 Motivation: The Need for Generalizable Spatial Perception

Traditional uncertainty models fail across multiple scenarios that are common in real-world robotics deployments. For instance, stereo cameras calibrated in indoor laboratory conditions produce overconfident uncertainty estimates when deployed in outdoor environments with varying lighting, leading to poor localization when integrated with other sensors. Similarly, IMU noise models tuned for one robotic platform (e.g., a ground vehicle) perform poorly when transferred to aerial platforms due to different vibration characteristics and mounting configurations.

Current robotic perception systems exhibit three fundamental limitations:

1. **Hand-Tuned Uncertainty Models:** Current perception systems rely on manually specified covariance matrices and noise parameters that assume stationary, Gaussian noise characteristics. These fixed models cannot capture the complex, environment-dependent variations in sensor reliability, leading to either overconfident or overly conservative estimates.
2. **Platform-Specific Calibration:** Each new robotic platform requires extensive manual parameter tuning by domain experts, involving weeks of data collection and empirical validation. This process does not scale to the diversity of robots and environments needed for widespread autonomous system deployment.
3. **Poor Cross-Modal Integration:** Existing multi-sensor fusion approaches treat uncertainties from different modalities as independent, ignoring the complex correlations that exist between visual, inertial, and other sensory measurements. This leads to suboptimal fusion weights and missed opportunities for uncertainty-driven sensor selection.

This dissertation addresses these limitations through a comprehensive learning-based uncertainty estimation framework that replaces hand-tuned models with data-driven approaches. By learning uncertainty models directly from sensor data across diverse conditions and platforms, we enable robotic perception systems that automatically adapt their reliability estimates, generalize across environments, and optimally fuse multi-modal information based on learned uncertainty correlations.

Chapter 2

Background

In this chapter, we establish the foundation for learning-based uncertainty estimation in robotic perception systems. In Section 2.1, we review the evolution from traditional sensor fusion to uncertainty-aware perception systems. In Section 2.2, we formalize the problem of generalizable uncertainty learning for spatial intelligence.

2.1 Related Works

2.1.1 Evolution from Sensor Fusion to Uncertainty Learning

Traditional robotic perception systems have relied on hand-tuned uncertainty models that assume fixed noise characteristics across different environments and platforms. Early probabilistic frameworks like the Extended Kalman Filter (EKF)[1] and Unscented Kalman Filter (UKF)[2] use manually specified covariance matrices for sensor measurements. While mathematically principled, these approaches struggle to capture the complex, environment-dependent nature of sensor noise, leading to suboptimal fusion performance and poor generalization across different operational conditions.

The recognition of this limitation has driven the evolution toward learning-based uncertainty estimation. Modern tightly coupled systems like VINS-Fusion [3] and OKVIS [4] still rely on factor graph optimization [5], but the fundamental challenge remains: how to accurately model uncertainty without extensive manual parameter tuning for each new environment or sensor configuration.

Recognizing the scalability and computational bottlenecks of tightly coupled methods, hybrid approaches emerged. A notable example is Super Odometry, which proposes an IMU-centric sensor fusion framework. By prioritizing the IMU as the primary estimator and selectively incorporating other sensor modalities, hybrid methods balance robustness, computational efficiency, and system simplicity—offering a more scalable solution compared to fully tightly coupled frameworks.

This gap has motivated the development of learning-based uncertainty estimation frameworks. Instead of relying on fixed covariance matrices, emerging approaches like VINet [6] and DynaSLAM [7] explore data-driven uncertainty models. However, most existing learning-based

methods focus on end-to-end state estimation rather than principled uncertainty modeling, limiting their theoretical foundations and practical deployment in safety-critical applications.

The emergence of foundation models represents a paradigm shift toward generalizable uncertainty estimation. Works like Dust3R [8] demonstrate that pre-trained models can provide robust uncertainty priors across diverse environments. This thesis extends this concept by developing specialized uncertainty learning frameworks for individual sensor modalities (visual, inertial) and their fusion, enabling truly generalizable spatial intelligence that adapts to new platforms and environments without manual recalibration.

Despite substantial progress, achieving robust, real-time, and scalable SLAM across diverse operational domains remains an active area of research.

2.1.2 Learning-Based Uncertainty Estimation

Capability	Visual-centric				LiDAR-centric					IMU-centric
	Droid-SLAM [9]	VINS-Mono[10]	OKVIS2.0[11]	Maplab2.0[11]	LVI-SAM[12]	Fast-LIO[13]	R3LIVE[14]	Fast-LVIO[15]	HF[16]	Our Thesis
Metrics-Aware Visual Uncertainty (C3)	×	×	×	×	×	×	×	×	×	✓
Learning Inertial Uncertainty Propagation (C4)	×	×	×	×	×	×	×	×	×	✓
Enhanced IMU Feature Observability (C5)	✓	×	×	×	×	×	×	×	×	✓
Cross-Modal Uncertainty Fusion (C7)	×	×	×	×	×	×	×	×	×	✓
Cross-Platform Uncertainty Adaptation (C8)	×	×	×	×	×	×	×	✓	✓	✓
Self-Supervised Uncertainty Calibration (C8)	×	×	×	×	×	×	×	×	×	✓
Contact-Aware Perception Models (C9)	×	×	×	×	×	×	×	×	×	✓
Uncertainty-Driven Locomotion Planning (C??)	×	×	×	×	×	×	×	×	×	✓
Physical Interaction Feedback Learning (C??)	×	×	×	×	×	×	×	×	×	✓
Generalizable Uncertainty Models (C8)	×	✓	×	×	×	×	✓	✓	✓	✓

Table 2.1: Non-complete overview of uncertainty learning capabilities across different robotic perception systems to highlight the under-explored research direction from prior works. Our thesis focuses on learning-based uncertainty estimation across individual sensors, multi-sensor fusion, and contact-rich robotics. ✓ indicates support, × indicates limited or no support.

2.1.3 Uncertainty-Aware Spatial Intelligence

The key challenge in developing truly intelligent robotic systems lies not just in sensor fusion, but in understanding and modeling the inherent uncertainty in perception. Real-world robotics applications—from underground exploration to contact-rich manipulation—expose sensors to conditions that deviate significantly from laboratory settings, making fixed uncertainty models inadequate.

Traditional approaches treat uncertainty as a nuisance to be minimized rather than valuable information to be leveraged. However, uncertainty quantification is fundamental to spatial intelligence: knowing what the robot doesn’t know enables safer decision-making, more robust planning, and adaptive behavior in uncertain environments. This paradigm shift from uncertainty avoidance to uncertainty awareness represents a critical evolution in robotic perception.

This thesis develops a comprehensive framework for learning-based uncertainty estimation across the entire robotic perception pipeline. Starting with individual sensor modalities (Chapter 3 for visual uncertainty, Chapter 4 for inertial uncertainty), progressing through multi-sensor fusion (Chapters 7 and 8), and extending to physical interaction scenarios (Chapters 9 and ??), we demonstrate that learned uncertainty models enable truly generalizable spatial intelligence.

2.2 Problem Definition

The goal of a sensor fusion SLAM (Simultaneous Localization and Mapping) system is to estimate the trajectory of a mobile robot and construct a consistent map of the surrounding environment by fusing complementary data from multiple heterogeneous sensors. These sensors typically include, but are not limited to, inertial measurement units (IMUs), cameras (monocular, stereo, or RGB-D), LiDAR, and thermal or radar sensors.

Formally, let $\mathcal{X}_{1:T} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ denote the sequence of robot poses over time, and let $\mathcal{M}$ denote the map of the environment. The robot receives a sequence of multi-modal sensor measurements $\mathcal{Z}_{1:T} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_T\}$, where each $\mathbf{z}_t$ may consist of visual, inertial, LiDAR, or other sensor observations at time t .

The core problem is to estimate the posterior:

$$p(\mathcal{X}_{1:T}, \mathcal{M} \mid \mathcal{Z}_{1:T}), \quad (2.1)$$

which is generally intractable and approximated through filtering, smoothing, or optimization techniques.

A sensor fusion SLAM system must address the following challenges:

- **Heterogeneous Sensor Characteristics:** Each sensor modality has different noise models, failure modes, and operating conditions, requiring careful calibration and uncertainty modeling.
- **Asynchronous and Multi-rate Data:** Sensors often operate at different frequencies and latencies, demanding accurate time synchronization and robust interpolation strategies.
- **Degenerate and Degraded Conditions:** The system must remain operational in environments where certain modalities fail, such as low-light for cameras or featureless regions for LiDAR.
- **Real-time Constraints:** To be practical for autonomous systems, sensor fusion and state estimation must be performed efficiently and within the constraints of onboard compute resources.

The objective is to design a SLAM framework that fuses multi-modal sensory information in a resilient and adaptive manner, enabling long-term, accurate localization and mapping in complex, dynamic, and degraded environments.

Part I

Part I: Metrics-Aware Uncertainty for Visual Odometry

*This part establishes metrics-aware uncertainty for odometry. We first develop visual uncertainty and 2D→3D covariance propagation in **MAC-VO**, then learn motion-adaptive IMU uncertainty in **AirIMU**, and finally enhance inertial odometry with uncertainty-aware models in **AirIO**.*

These chapters replace hand-tuned noise with learned, scale-consistent covariance for robust two-frame optimization, setting up multi-modal fusion next.

Chapter 3

MAC-VO: Metrics-Aware Covariance for Stereo Visual Odometry

3.1 Introduction

Visual Odometry (VO) predicts the relative camera pose from image sequences and often serves as the front-end of Simultaneous Localization and Mapping (SLAM) systems. Over the past few decades, both geometric and learning-based methods have been developed with significant advances in generalizability and accuracy [9, 17–19]. However, VO remains a challenging problem in real-world scenarios, with multiple visually degraded scenarios such as low illumination, dynamic and texture-less scenes.

To improve the robustness in challenging scenes, geometric-based VO algorithms employ outlier filtering strategies [20] and weigh the optimization residuals by the covariance of the observed features [21]. However, how to effectively *select the reliable keypoints* and *model their covariance* becomes two significant challenges. Existing methods typically select the keypoints based on local intensity gradient with a manually defined threshold [22–24]. These approaches lead to errors and outliers because they don’t model the structure or context information of the environment (e.g., features on repetitive patterns may not be ideal candidates despite high image gradients). Moreover, the covariance model is often empirically modeled using a constant parameter, which is sub-optimal. In addition, the parameters in the keypoint selection and covariance model need to be extensively tuned for different environments.

With advances in learning-based visual features, more algorithms utilize learned features [22] to optimize the camera pose. Confidence score [25] or confidence weights [9, 19] of these feature points are often obtained in an unsupervised manner. The learned confidence helps to track the features and model the reliability during the optimization. However, these confidence or uncertainty values are scale-agnostic, which means they don’t reflect the actual estimation error in the 3D space. This scale-agnostic problem brings two limitations. Firstly, it makes the covariance inconsistent across different environments that vary in scale, such as indoors and outdoors. Secondly, it makes it harder to integrate multiple constraints from different modalities or sensors.

To overcome the above challenges, this paper addresses the problem of *modeling metrics-aware covariance values for the 3D keypoints*. More specifically, this is achieved through two innovations.

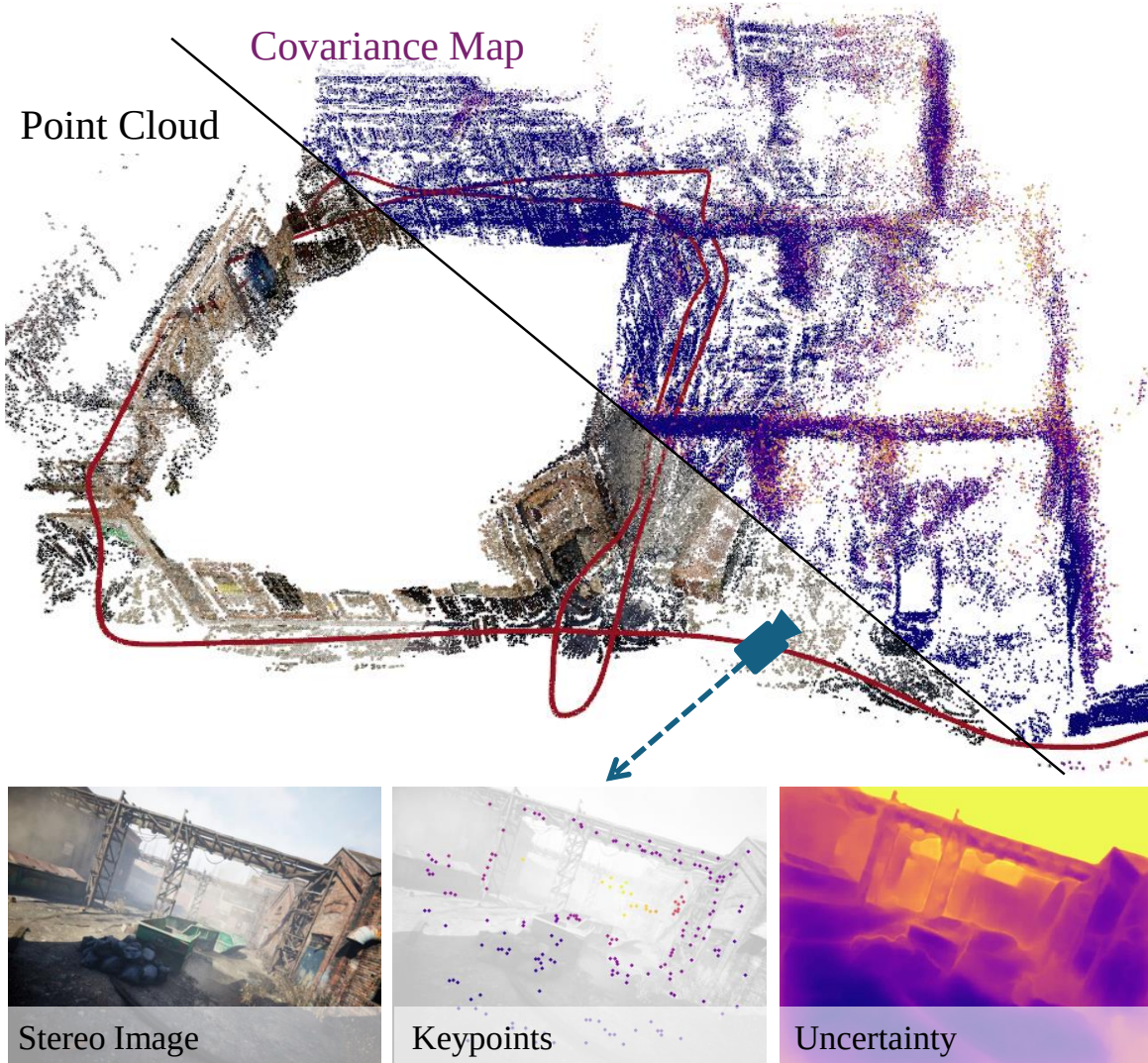


Figure 3.1: Without relying on multi-frame bundle adjustment, MAC-VO aggregately reconstructs the map based on the two-frame Pose graph optimization. We propose a *metrics-aware covariance* model for 3D keypoints based on learned matching uncertainty.

Firstly, we propose a learning-based model to quantify the 2D metrics-aware uncertainty of feature matching. Inspired by the FlowFormer [26] and GMA [27], we employ an iterative update model and motion aggregator to predict the uncertainty in 2D image space, which helps to filter unreliable features in the occluded region or low-illumination area. Secondly, based on the learned 2D uncertainty values, we model the covariance of the feature points in 3D space using a *metrics-aware 3D covariance model*. Compared to DROID-SLAM [9], which utilizes a scale-agnostic diagonal covariance matrix, our approach provides a more accurate representation by modeling the covariance of 3D feature points. This covariance model includes the inter-axes correlation of the 3D features. In the ablation study, we demonstrate the inter-frame consistency and the intra-frame consistency of our proposed covariance model.

We integrate the above two innovations into MAC-VO, a stereo VO that features superior keypoint selection and pose graph optimization based on the metrics-aware covariance model and achieves accurate tracking results in challenging cases compared with state-of-the-art VOs and

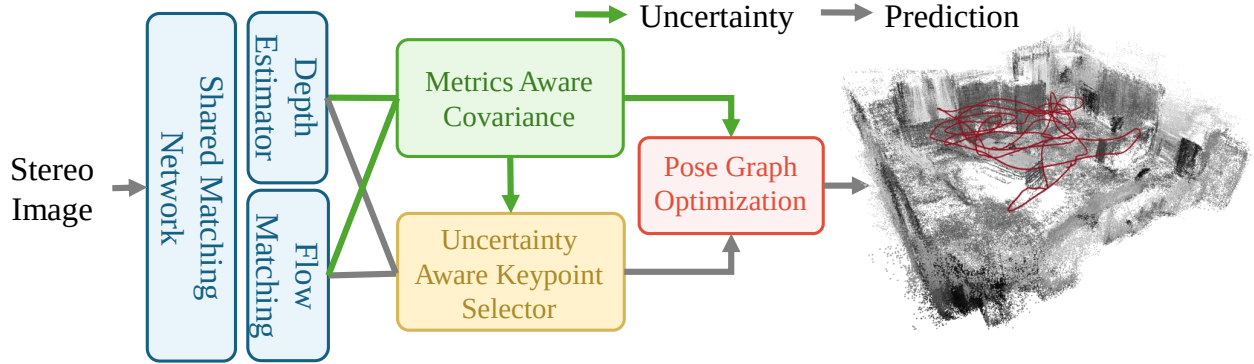


Figure 3.2: MAC-VO pipeline. It employs a shared matching network to jointly estimate depth, optical flow, and corresponding uncertainties. The learned uncertainty is utilized in the keypoint selector, the metrics-aware 3D covariance model, and the back-end optimization.

even some SLAM systems without fine-tuning or multi-frame optimization. In summary, the main contributions are:

- We present a learning-based 2D uncertainty network with metrics awareness, leveraging an iterative motion aggregator to capture the inconsistency of the feature matching. We use this metrics-aware uncertainty to evaluate the quality of features in order to select keypoints and guide the back-end optimization.
- This paper introduces a novel metrics-aware 3D covariance model based on the 2D uncertainty of feature matching and depth estimation. The ablation study demonstrates the necessity of scale consistency and the off-diagonal terms in the pose graph optimization.
- We propose the MAC-VO, a stereo VO pipeline that estimates the camera pose and registers 3D features with metrics-aware covariance. In the experiments, MAC-VO outperforms existing VO algorithms and even some SLAM algorithms in challenging environments.

3.2 Related Works

Existing geometric-based methods optimize the camera pose based on geometric constraints like re-projection error [17, 28] or photometric error [29–31]. To more accurately model the uncertainty of the depth, Civera et al. [32] and Montiel[33] investigate the inverse depth parameterization. These methods often use constant parameters and simple heuristics to model the covariance matrix of these errors during the factor graph optimization. For multi-sensor fusion [3, 21] and semantic SLAM [34, 35], the covariance model plays a significant role in weighting the confidence of different sensors and modules. To capture sensor uncertainty, the covariance models are often tuned based on empirical prior. However, these simplified covariance models fail to capture the complexity of the challenging environments.

Recent advances in deep learning have transformed research in optical flow estimation [36, 37], feature matching [25, 38], depth estimation [39–41], and end-to-end camera pose estimation [18, 42]. Several methodologies have been developed to address the uncertainty in estimating depth, flow, and pose. Dexheimer et al. [43] proposed a learned depth covariance function that is applied in downstream tasks like 3D reconstruction. Nie et al. utilize a self-supervised learning

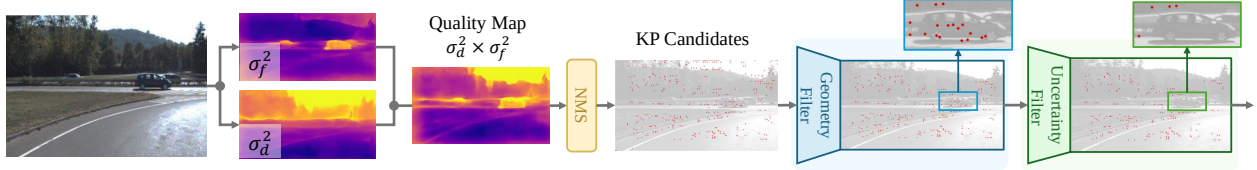


Figure 3.3: We include three filters: Non-minimum Suppression (NMS) filter, geometric filter, and uncertainty-based filter. In the KITTI dataset, the uncertainty filter implicitly filters out the unreliable feature matchings in the scene.

method to jointly train depth and depth covariance of images in the wild [44]. ProbFlow [45] proposes using post-hoc confidence measures to assess the per-pixel reliability of flow.

Amidst these developments, hybrid learning-based SLAM systems [40, 46–50] are emerging to synergize the geometric constraint with the adaptability of learning-based methods. To improve the reliability of the feature tracking process, some methods introduce learning-based uncertainty measurements or confidence scores [25, 51–54] for pose optimization. DROID-SLAM [9] utilizes a differentiable bundle adjustment layer to implicitly tune the uncertainty model. This method employs a simplified diagonal covariance model for the bundle adjustment. These methods focus on the relative confidence between features, but ignore the scale consistency of the covariance model.

For keypoint selection, geometric-based VO relies on hand-crafted features [23, 24], which detect the edges and corner points. Recent advances in learning-based methods train feature extractors in a data-driven manner [22, 55]. However, these keypoints also prioritize edge and corner features due to data bias in the pre-training dataset. Recent works like D3VO [46] have shown that relying on edge and corner points can degrade state estimation, sometimes performing worse than random selectors [19]. The accuracy of learning-based feature matching and depth estimation algorithms is particularly compromised at object edges due to feature interpolation and the ambiguity in neural networks. In this work, we propose a keypoint selector based on learned uncertainty to filter out the unreliable features.

3.3 Method

As illustrated in Figure 3.2, MAC-VO outlines an effective integration of a learning-based front-end and a geometrically constrained back-end using the metrics-aware covariance model. In the front-end (Section 3.3.1), we train an uncertainty-aware matching network to model the corresponding uncertainty that stems from feature deficiencies. Utilizing the learned uncertainty, we develop a keypoint selector in Section 3.3.2 to choose reliable features. The metrics-aware 2D uncertainty is then propagated to the 3D space via the proposed covariance model in Sec. 3.3.3. In the back-end optimization (Sec. 3.3.4), we optimize the relative motion by minimizing the distance between registered keypoints weighted by the 3D covariance.

3.3.1 Network & Uncertainty Training

The objective of the proposed matching network is to estimate the flow $\hat{f} \in \mathbb{R}^2$ and the corresponding uncertainty $\hat{\Sigma}_f = \text{diag}(\sigma_u^2, \sigma_v^2)$ between two frames. We adopt the FlowFormer architecture [26] as the backbone for optical flow estimation, leveraging its transformer-based

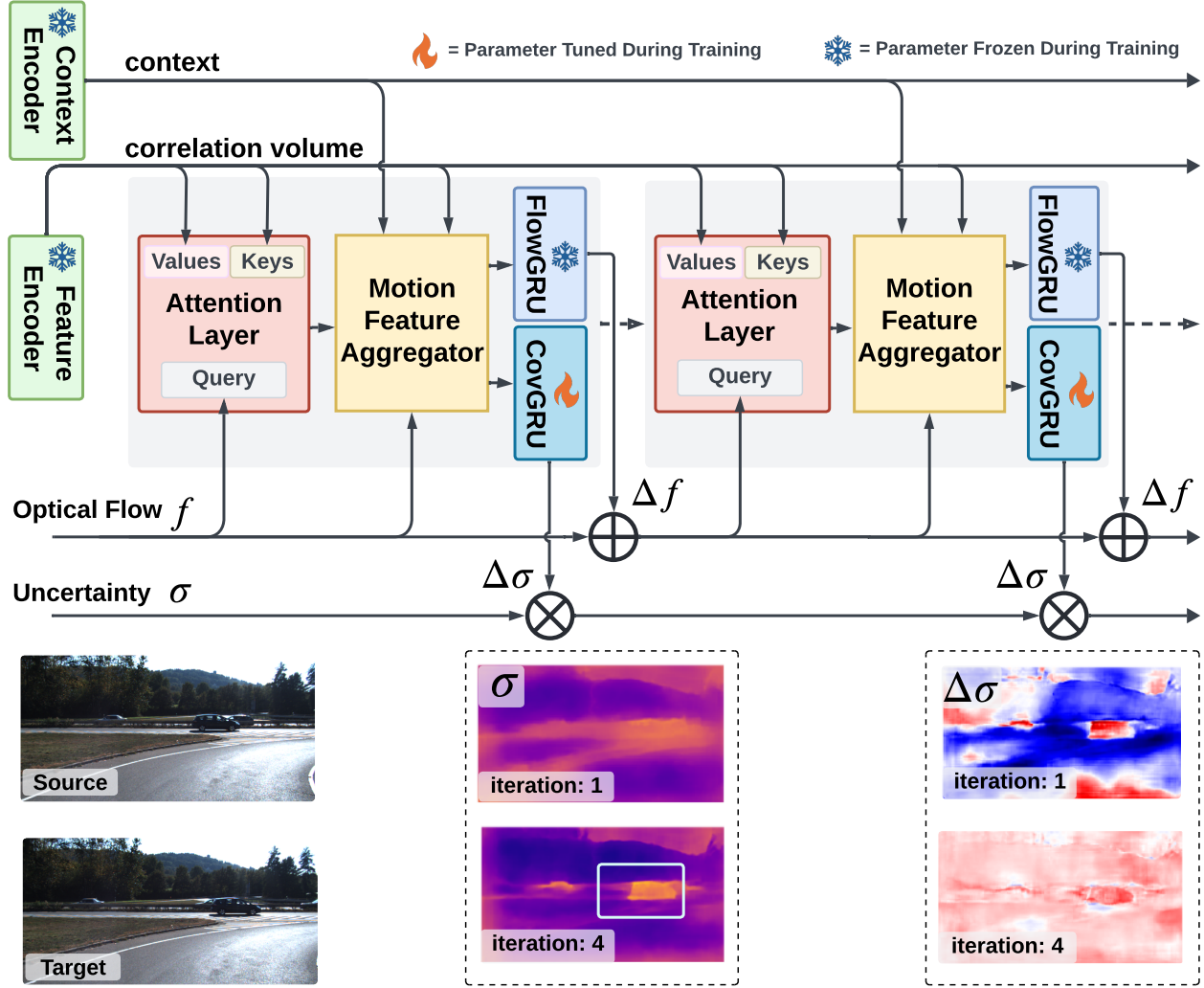


Figure 3.4: Top: Architecture of the uncertainty-aware matching network. **Bottom:** Each iteration, the model captures the inconsistency of the matching. For the $\Delta \sigma$, the red color indicates a positive $\Delta \sigma$ increasing the uncertainty, and blue means decreasing uncertainty.

design. As shown in Figure 3.4, the recurrent decoder iteratively refines flow predictions by capturing ambiguities in feature correspondence within the cost volume’s matching space, utilizing global context from the encoded cost memory.

To extend this framework and leverage both global motion cues and local features, we introduce a covariance decoder that predicts $\log \Delta \sigma$, the iterative updates of the uncertainty in log space. Operating the uncertainty update in log space facilitates additive updates, ensures positive variances, and stabilizes gradients. After iterative updates, the log-variance passes through an $\exp$ activation function to obtain the final uncertainty.

To supervise the uncertainty, we leverage the negative log-likelihood loss used in conformal prediction [56–58]:

$$L_{cov} = \sum_i^N \alpha_i \left((y - \hat{f}_i)^\top \hat{\Sigma}_f^{-1} (y - \hat{f}_i) + \log(\det \hat{\Sigma}_f) \right), \quad (3.1)$$

where y is the ground truth optical flow, f_i denotes the i -th iteration of the network outputs, and α_i is the weight for each iteration and is set to decrease exponentially with a ratio of 0.8. During the training stage, we initialize the encoder network parameters with the pre-trained model by FlowFormer. We then train the covariance module on the synthetic dataset TartanAir [59], and evaluate the model in the TartanAirv2 sequences and other real-world datasets. Our experiments demonstrate that the model can generalize to real-world datasets without fine-tuning.

3.3.2 Uncertainty-based Keypoint Selection

Unlike the random selector used in DPVO[19] and the hand-crafted feature selector used in ORB-SLAM [23], we leverage the learned uncertainty estimation to filter out unreliable features. This is achieved by composing three filters: the *uncertainty filter*, *geometry filter*, and the *non-maximum suppression* (NMS). To prevent the clustering of keypoint candidates, we first apply the NMS filter, ensuring a more even spatial distribution of features. The geometry filter further removes keypoints located at the image boundaries and those outside the valid depth observation range. Finally, the uncertainty filter eliminates pixels with depth and flow uncertainty greater than 1.5 times the median uncertainty of the current frame. Illustrated in Figure 3.3, the uncertainty filter removes all keypoint candidates on the moving vehicle on a KITTI trajectory. Our proposed uncertainty estimation strategy also filters out keypoints on occluded objects, reflective surfaces, and feature-less areas, resulting in a more robust and reliable feature set.

3.3.3 Metrics-aware 3D Covariance Model

In the context of the camera projection geometry, the covariance of a 3D keypoint is determined by the uncertainty of the depth σ_d^2 and matching (σ_u^2, σ_v^2). To accurately model the covariance for 3D keypoints, it is critical to determine (1) the depth uncertainty of the matched points $\hat{\sigma}_d^2$ and (2) the off-diagonal covariance terms during the 2D-3D projection.

Project 2D Uncertainty to 3D Covariance Following the pinhole camera model with focal length f_x, f_y and optical center c_x, c_y , the coordinate of the keypoint is calculated by: $\mathbf{x}_{i,t} = (\mathbf{u}_{i,t} - c_x)\mathbf{d}_{i,t}/f_x$, $\mathbf{y}_{i,t} = (\mathbf{v}_{i,t} - c_y)\mathbf{d}_{i,t}/f_y$ and $\mathbf{z}_{i,t} = \mathbf{d}_{i,t}$, as shown in Figure 3.5 (a). To accurately capture the uncertainties associated with these measurements, the main diagonal of

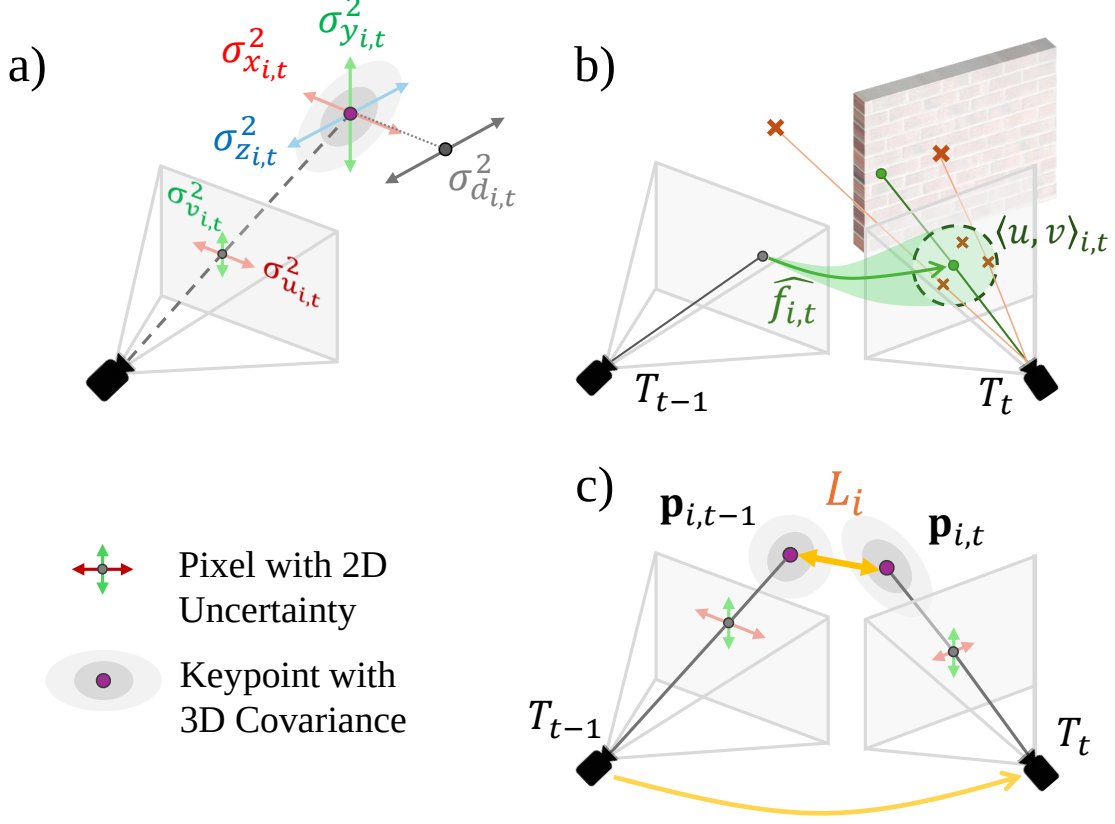


Figure 3.5: **a)** Uncertain estimation of depth due to the error in feature matching. **b)** Projecting depth and matching uncertainty from image plane to 3D space. **c)** Residual $\mathcal{L}_i$ for pose graph optimization.

the covariance matrix is formulated as:

$$\begin{aligned}\sigma_{x_{i,t}}^2 &= \left(\sigma_{u_{i,t}}^2 \sigma_{d_{i,t}}^2 + \sigma_{u_{i,t}}^2 d_{i,t}^2 + (u_{i,t} - c_x)^2 \sigma_{d_{i,t}}^2 \right) / f_x^2, \\ \sigma_{y_{i,t}}^2 &= \left(\sigma_{v_{i,t}}^2 \sigma_{d_{i,t}}^2 + \sigma_{v_{i,t}}^2 d_{i,t}^2 + (v_{i,t} - c_y)^2 \sigma_{d_{i,t}}^2 \right) / f_y^2, \\ \sigma_{z_{i,t}}^2 &= \sigma_{d_{i,t}}^2.\end{aligned}\tag{3.2}$$

In this model, the projected coordinates are interdependent due to the common multiplier of depth $d_{i,t}$. To precisely formulate the covariance of the 3D keypoints ${}^c\Sigma_{i,t}^p$ under camera coordinate, it is essential to include the off-diagonal covariance terms in ${}^c\Sigma_{i,t}^p$.

$${}^c\Sigma_{i,t}^p = \begin{bmatrix} \sigma_z^2 & \sigma_{xz_{i,t}} & \sigma_{yz_{i,t}} \\ \sigma_{xz_{i,t}} & \sigma_{x_{i,t}}^2 & \sigma_{xy_{i,t}} \\ \sigma_{yz_{i,t}} & \sigma_{xy_{i,t}} & \sigma_{y_{i,t}}^2 \end{bmatrix} \begin{aligned} \sigma_{xz} &= \sigma_d^2 (u - c_x) / f_x \\ \sigma_{yz} &= \sigma_d^2 (v - c_y) / f_y \\ \sigma_{xy} &= \frac{\sigma_d^2 (u - c_x)(v - c_y)}{f_x f_y} \end{aligned}\tag{3.3}$$

Results in the ablation study also confirm the necessity of off-diagonal terms for accurate pose graph optimization. Detailed derivation for Eq. 3.2 and Eq. 3.3 is in Appendix ??.

Uncertainty Correction after Keypoint Matching As shown in Figure 3.5 (b), the

matched features are expected to fall in a probabilistic range centered at $[u_{i,t}, v_{i,t}]$ with the flow uncertainty $(\sigma_{u_{i,t}}^2, \sigma_{v_{i,t}}^2)$. However, the depth observations of matched keypoints may vary due to scene geometry. For example, if a matched keypoint is located near the edge of an object, such as a wall, even a minor perturbation in feature matching can lead to substantial discrepancies in 3D keypoint registration. This, in turn, amplifies the uncertainty in the reconstructed 3D structure.

To address this problem, we correct the depth uncertainty $\hat{\sigma}_{d_{i,t}}^2$ of the matched feature point based on the depth feature of the local patch $D_{i,t}$ with the kernel size of 32. We approximate it with the weighted sum of the variances within the patch. The weights are determined by a 2D Gaussian kernel φ , which utilizes $\sigma_{u_{i,t}}^2$ and $\sigma_{v_{i,t}}^2$ to adjust the influence of each point within the patch: $\sigma_{d_{i,t}}^2 = \sum_j \varphi_j (d_j - \mu_{D_{i,t}})^2$.

3.3.4 Pose Graph Optimization

We optimize the camera pose $T_t \in SE(3)$ at time t in the world frame by minimizing the distance of the matched 3D keypoints $p_{i,t-1}$ and ${}^c p_{i,t}$, where ${}^c p_{i,t}$ is in the camera frame. To reduce the initial error margin of the optimization, we initialize the camera pose using the relative motion estimated by the TartanVO [18].

The pose graph optimization is formulated as follows:

$$\begin{aligned} T^* &= \arg \min_{T_t} \sum_i \| \mathbf{p}_{i,t-1} - T_t {}^c \mathbf{p}_{i,t} \|_{\Sigma_i}^2, \\ \Sigma_i &= \Sigma_{i,t-1}^p + R_t {}^c \Sigma_{i,t}^p R_t^\top. \end{aligned} \tag{3.4}$$

$\| \cdot \|_{\Sigma_i}$ represents the Mahalanobis distance with covariance matrix Σ_i . Unlike DROID-SLAM [9], which employs a diagonal covariance matrix, we model the correlation between axes to capture accurate inter-dependencies. We solve this problem using Levenberg-Marquardt optimization via *PyPose* [60, 61].

3.4 Experiment

Datasets & Baseline We evaluate the proposed model and baseline methods on a variety of public datasets, including synthetic dataset TartanAir v2 [59], real-world data from EuRoC [62], KITTI [63], as well as customized data collected from a Zed-X camera. These datasets cover a diverse range of hardware configurations, motion patterns, and environments. To demonstrate our method’s robustness under challenging scenarios, we collected the TartanAir v2, a new set of difficult trajectories following the TartanAir [59] that includes frequent indoor-outdoor transition and low-illumination scenes as shown in Figure 3.6. To demonstrate the generalizability of our model, we use the same configuration and same matching model across all datasets.

Evaluation Metrics Since the proposed method does not contain loop closure or global bundle adjustment, our evaluation focuses on relative error. We use relative translation error

(t_{rel} , m/frame) and rotation error (r_{rel} , °/frame) as:

$$t_{\text{rel}} = \frac{1}{N} \sum_{t=1}^N \left\| \mathbf{p}_{t+1} - \mathbf{p}_t - R_t \hat{R}_t^\top (\hat{\mathbf{p}}_{t+1} - \hat{\mathbf{p}}_t) \right\|_2,$$

$$r_{\text{rel}} = \frac{180}{\pi} \frac{1}{N} \sum_{t=1}^N \left\| \log \left(\hat{R}_{t,t+1}^\top R_{t,t+1} \right) \right\|_2,$$
(3.5)

where $\mathbf{p}_t$ and R_t are ground truth position and rotation, $\hat{\mathbf{p}}_t$ and $\hat{R}_t$ is the estimated position and rotation. $R_{t,t+1} = R_t^\top R_{t+1}$ is the rotation from frame t to frame $t+1$.

3.4.1 Quantitative Analysis

Table 3.1: Performance comparison of different methods on the EuRoC Dataset. Only odd-ordered trajectory is shown due to page limit, see Appendix ?? for the remaining results. Average is calculated over all trajectories of EuRoC.

Trajectory	MH01		MH03		MH05		V102		V201		V203		Avg.	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
SLAM														
 ORB-SLAM 3	0.0035	0.0450	0.0058	0.0603	0.0059	0.0526	0.0096	0.1757	0.0064	0.1615	0.033	0.9497	0.0092	0.1866
 DROID-SLAM*	0.0012	0.0159	0.0034	0.2656	0.0025	0.0193	0.0026	0.0417	0.0012	0.0289	0.0034	0.1033	0.0024	0.0590
 MAST3R-SLAM*	0.0143	0.5493	0.0493	0.8890	0.0254	0.5808	0.0354	1.3178	0.0117	0.6818	0.0430	2.3306	0.0277	1.0472
VO														
 TartanVO*	0.0121	0.0560	0.0302	0.2791	0.0193	0.0604	0.0251	0.1244	0.0065	0.0920	0.0303	0.2986	0.0198	0.1270
 TartanVO	0.0277	0.5122	0.0514	0.6635	0.0464	0.4797	0.0394	1.0420	0.0195	0.4684	0.0473	1.9657	0.0368	0.8346
 iSLAM-VO	0.0042	0.0560	0.0076	0.2789	0.0070	0.0603	0.0066	0.1241	0.004	0.0920	0.0151	0.2984	0.0071	0.1269
 DPVO*	0.0015	<u>0.0207</u>	<u>0.0028</u>	<u>0.0273</u>	<u>0.0028</u>	0.0243	0.0041	0.0496	<u>0.0016</u>	<u>0.0342</u>	<u>0.0045</u>	<u>0.1205</u>	<u>0.0027</u>	<u>0.0437</u>
 Ours	<u>0.0014</u>	0.0214	0.0023	0.0238	0.0025	<u>0.0216</u>	<u>0.0029</u>	<u>0.0434</u>	0.0012	0.0289	0.0049	0.1284	0.0024	0.0403

 Monocular method.  Stereo method. * Scale-aligned with ground truth.

EuRoC Dataset We assessed our model on the EuRoC [62] dataset, as detailed in Table 3.1, comparing it against baseline methods including visual odometry and state-of-the-art visual SLAM systems with loop-closure and global bundle adjustment. While our method exhibits compatible performance to DROID-SLAM on average t_{rel} , it outperforms all baselines in terms of r_{rel} by around 10%.

Table 3.2: Performance comparison on the TartanAir v2 Hard Dataset. Noticeably, ORB-SLAM3 and MAST3R-SLAM lost track on all sequences and are not included. See Appendix ?? for the *Easy* subset. Average is calculated over all trajectories of TartanAir v2 Hard.

Trajectory	H00		H01		H02		H03		H04		H05		H06		Avg.	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
SLAM																
 DROID-SLAM*	<u>.0485</u>	<u>.1174</u>	.0023	.0210	<u>.0190</u>	<u>.0821</u>	.0064	<u>.0300</u>	.0057	.0255	.1463	<u>.2357</u>	<u>.0310</u>	.0908	<u>.0370</u>	.0861
VO																
 TartanVO*	.1605	3.338	.2918	2.775	.2718	3.305	.2775	2.191	.2204	2.874	.1644	2.899	.2350	3.756	.2316	3.020
 TartanVO-stereo	.1505	.4329	.0914	.7542	.0715	.3265	.0842	.4053	.0678	.6569	<u>.0803</u>	1.186	.0784	.9458	.0892	.6725
 iSLAM-VO	.4235	2.630	.3070	3.018	.3252	2.189	.3622	2.435	.2576	2.899	.2574	3.755	.2099	3.145	.3061	2.867
 DPVO*	.4984	.4937	.1738	.7112	.0539	.2539	.3847	2.703	.0481	.1869	.1891	1.430	.3365	2.943	.2406	1.246
 Ours	.0085	.1018	<u>.0344</u>	<u>.1450</u>	.0048	.0628	<u>.0150</u>	<u>.0778</u>	<u>.0092</u>	<u>.1414</u>	.0048	.0552	.0217	<u>.4160</u>	.0141	<u>.1429</u>

 Monocular method.  Stereo method. * Scale-aligned with ground truth.

TartanAir v2 Dataset TartanAir v2 presents significant challenges for visual odometry due

Table 3.3: Performance comparison of different methods on the KITTI Dataset. Only even numbered trajectory is shown due to page limit, see Appendix ?? for remaining results. Average is calculated over all trajectories of KITTI Odometry.

Trajectory	00		02		04		06		08		10		Avg.	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
SLAM														
 ORB-SLAM 3	0.0252	0.0586	0.0438	0.0529	0.0274	0.0322	0.0228	0.0338	<u>0.0271</u>	0.0455	0.0166	<u>0.0515</u>	0.0258	<u>0.0434</u>
 DROID-SLAM*	<u>0.0198</u>	<u>0.0538</u>	<u>0.0250</u>	<u>0.0445</u>	<u>0.0255</u>	<u>0.0296</u>	<u>0.0199</u>	<u>0.0296</u>	0.0275	<u>0.0381</u>	0.0309	0.0715	0.0900	0.0448
 MAST3R-SLAM*	0.8988	0.7154	1.0104	0.5959	1.2266	0.1124	1.2599	0.1289	0.7760	0.6075	0.6889	0.4709	0.9538	0.4202
VO														
 TartanVO*	0.2066	0.1055	0.1626	0.1105	0.1152	0.0789	0.2234	0.0816	0.1857	0.0823	0.1745	0.0907	0.2207	0.0886
 TartanVO	0.0656	0.1026	0.0905	0.1197	0.1747	0.1158	0.0923	0.0968	0.0721	0.1063	0.0679	0.0969	0.1804	0.1147
 iSLAM-VO	0.0577	0.1052	0.0686	0.1101	0.1356	0.0787	0.0837	0.0812	0.0510	0.082	0.0449	0.0905	0.0878	0.0883
 DPVO*	0.4542	0.0495	0.4209	0.0381	0.0348	0.0219	0.2393	0.0250	0.3051	0.0347	0.0661	0.0386	0.1951	0.0329
 Ours	0.0192	0.0654	0.0223	0.0715	0.0206	0.0473	0.0187	0.0456	0.0254	0.0509	<u>0.019</u>	0.0569	<u>0.0420</u>	0.0645
 Monocular method.  Stereo method. * Scale-aligned with ground truth.														

to its rigorous motion and extreme illumination changes. The traditional method, ORB-SLAM, fails on most trajectories. Our approach improves 61.9% in t_{rel} compared to DROID-SLAM. Notably, on trajectory H00, which simulates the lunar surface shown in Figure 3.6, our model demonstrates a remarkable 82.4% decrease in t_{rel} and achieves the lowest r_{rel} among all baseline methods.

KITTI Dataset To further validate our robustness and consistency in outdoor, large-scale trajectories with dynamic objects, we evaluate our system on the KITTI [63] dataset. Our method, which relies solely on two-frame pose optimization, shows commendable performance, ranking behind ORB-SLAM3, a full visual SLAM system, on t_{rel} . Our system significantly outperforms other visual odometry approaches in t_{rel} , demonstrating a 53.3% reduction in relative translation error. The degraded performance of r_{rel} may be attributed to the lack of multi-frame optimization.

3.4.2 Qualitative Analysis and Ablation Study

In addition to quantitative evaluations, we conduct qualitative evaluations of our proposed system across multiple datasets including EuRoC, KITTI, and TartanAir v2, supplemented by manually collected data using a ZED-X stereo camera. MAST3R-SLAM struggles on many trajectories of KITTI and TartanAir, primarily due to incorrect loop closure detections from visual ambiguities. Our model, even without multi-frame optimization, achieves top-tier performance and exhibits fewer glitches than baseline methods. As demonstrated in the Figure 3.6, our method produces smoother trajectories and superior pose estimation precision. Figure 3.6 a) presents that our model correctly identifies the region occluded by the mounted platform and dynamic objects in the scene and assigns a high uncertainty score to these regions.

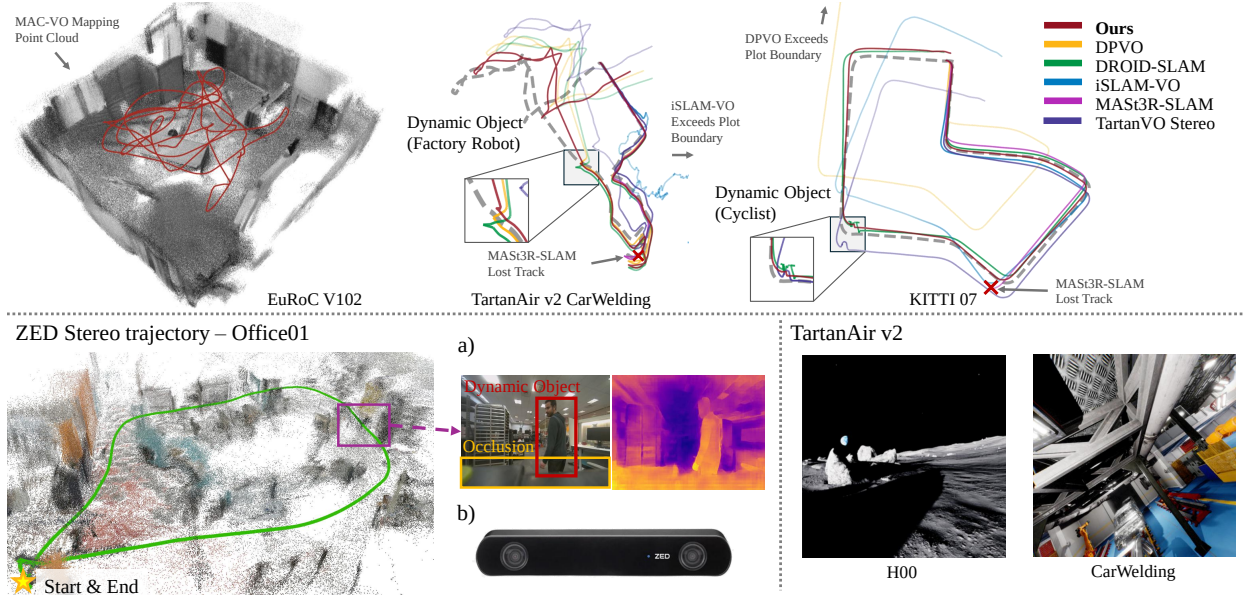


Figure 3.6: **Top:** Trajectories estimated by our model and baselines. We highlight the segments where dynamic objects interrupt the VO. **Bottom left:** We collect data in the office using our own payload with the ZED-X camera. The data contains multiple dynamic objects and visual occlusions. **Bottom right:** samples of TartanAir v2 test dataset, which simulates the exotic lunar environment.

Table 3.4: Ablation study. Detailed results in Appendix ??.

Dataset	TartanAir v2 Hard		TartanAir v2 Easy	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}
Module Ablation				
w/o CovKP & CovOpt	.0743	.5221	.0521	.2799
w/o CovOpt	.0679	.3776	.0511	.2367
w/o CovKP	<u>.0188</u>	.2347	<u>.0066</u>	.0808
Covariance Ablation				
DiagCov	.0461	.3023	.0277	.1548
Scale Agnostic	.0204	<u>.2321</u>	.0086	<u>.0764</u>
Ours	.0141	.1429	.0051	.0670

Table 3.5: Time spent on each module of MAC-VO under different optimization with image resolution of 640×640 .

	Raw	TRT [*]	MP [†] + TRT [*]	MAC-VO Fast [‡]
Modules (ms)				
Frontend Network	401.9	239.3	239.9	81.6
Optimization	53.4	57.5	-	-
Motion Model	5.2	5.1	5.3	5.2
Keypoint Selector	0.7	0.6	0.6	0.5
Covariance Model	0.6	0.7	0.7	0.7
Overall (fps)	2.15	3.25	3.96	10.57

* TRT: TensorRT framework - <https://developer.nvidia.com/tensorrt>

† MP: multi-processing the PGO in parallel with the matching network.

‡ MAC-VO Fast: utilizes half-precision number (bf16) and light-weight model.

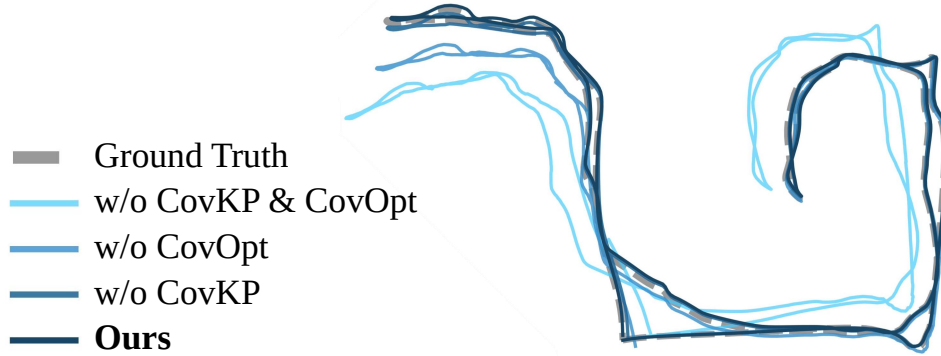


Figure 3.7: Results of various ablation setups on the H01 of TartanAir v2.

Ablation Study In Table 3.4, we first perform the ablation study on each module of MAC-VO including (a. *w/o CovKP & CovOPT*) with random keypoint selector and identity covariance matrix. (b. *w/o CovOpt*) replace the metrics-aware covariance model with the identity covariance model. (c. *w/o CovKP*) replace the proposed keypoint selector with a random keypoint selector. To demonstrate the necessity of scale consistency and off-diagonal terms in our covariance model, we run the ablation study with different configurations: (I. *DiagCov*) remove off-diagonal terms in Eq. 3.3, (II. *Scale-agnostic*) normalize the covariance model by the average determinants of covariance matrices of each frame.

3.4.3 Runtime Analysis

The runtime analysis shown in Table 3.5 uses the platform with AMD Ryzen 9 5950X CPU and NVIDIA 3090 Ti GPU. We also introduce a fast mode (MAC-VO Fast) that utilizes half-precision number in the network inference to enhance efficiency. This mode also speeds up the memory decoder network by reducing the number of iterative updates from 12 to 4. The fast mode performs at 10.5 fps (frames per second) with 70% of the performance of the original MAC-VO. More details are included in Appendix ??.

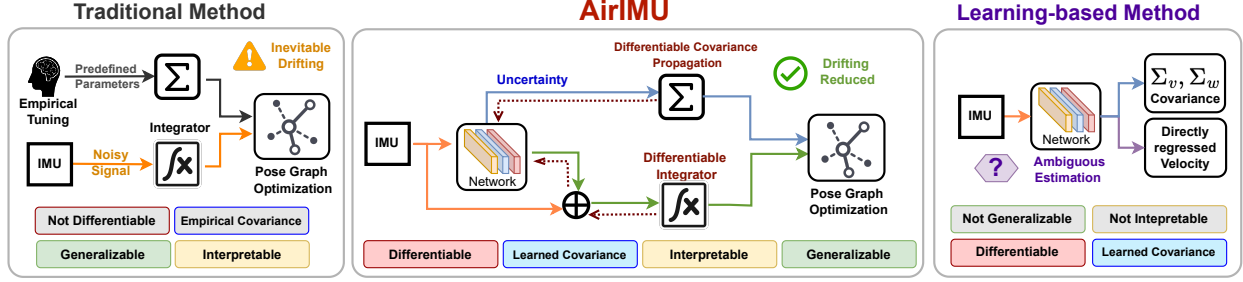


Figure 4.1: **Left:** Traditional methods with empirically set uncertainty parameters often fail to capture noise inherent in the sensor accurately. **Middle:** AirIMU corrects the IMU noise and predicts the corresponding uncertainty of the integration. Notably, its covariance propagation is differentiable, promoting efficient training. **Right:** Learning-based methods predict velocity and the corresponding covariance but lack interpretability and generalizability across different data domains.

Chapter 4

AirIMU: Learning Uncertainty Propagation for Inertial Odometry

4.1 Introduction

Inertial Measurement Unit (IMU), capable of sensing high-frequency linear acceleration and angular velocity, is an essential component in almost all kinds of robots [64–66]. As a result, inertial odometry (IO), which aims to estimate the relative movements of robots using IMU measurements, has broad applications in autonomous driving [66], navigation [67], augmented reality [68], and human motion tracking [69]. Due to the IMUs’ relative independence from the external environment, IO is insensitive to environmental disturbances such as illumination changes and visual obstructions [64]. Therefore, IO has great potential in robust state estimation under complex operation conditions, especially compared to exteroceptive sensors like cameras and LiDAR [3]. Nevertheless, the performance of existing IO algorithms, either the model-based [70] or the data-driven methods [71], is still unsatisfactory for real-world applications [64].

Kinematic model-based IO methods have excellent generalizability but often suffer from complex and multifaceted noise [72]. The IMU sensor noises can be classified into two categories: the deterministic errors and the non-deterministic errors [73]. The deterministic errors are the

systematic biases in the measurements that can be consistently predicted or modeled, often arising from factors such as installation imperfections, and calibration errors [74]. Conversely, the non-deterministic errors, which are also known as stochastic errors, are related to factors like mechanical shocks, vibration influences, thermal effects, mechanical defects, and calculation ambiguities [75–77]. Many methods employ linearized models to handle the deterministic errors by accounting for additive sensor bias, scale factors, and axis misalignment [78–80] and handle non-deterministic errors using zero-mean Gaussian function [3, 73]. However, these simplified linear models struggle to fully account for non-deterministic uncertainties from the natural environments, highlighting a critical area for improvement. Data-driven IO, which use neural networks to predict the motion and implicitly fit the IMU noise, often face challenges with interpretability and generalizability. To be more specific, these methods often overfit in specific motion patterns such as pedestrian [81] or vehicle [82], and fail to generalize to scenarios outside of the training distribution. For example, the model trained on hand-held platforms running on flat ground fails to generalize to robots climbing stairs or drones [83]. Moreover, replacing the motion models with the neural networks undermines the model’s interpretability.

Recent advances in hybrid methods show the potential to merge the benefits of both the model-based method and the data-driven method. For example, some researchers fuse the data-driven motion model with the IMU kinematic motion model using EKF [84, 85] or batched optimization [86]. These solutions improve the robustness of the state estimation but still struggle with generalizability. On the other hand, efforts have been made in training neural networks to model the error in gyroscope [82], accelerometer [87], IMU bias evolution [83]. Despite these advancements, none of these algorithms explicitly model the uncertainty of the IMU measurements in the IMU pre-integration. When deploying the data-driven model with visual-inertial odometry [83, 87], these algorithms still assume the IMU covariance model as a manually calibrated stochastic variable. These gaps hinder the deployment of hybrid methods for real-world robots.

To close these gaps, we present a hybrid method AirIMU as illustrated in Figure 4.3. It leverages the data-driven method to model the non-deterministic noise and the kinematic motion model to ensure generalizability in novel environments. To empower the AirIMU with the ability to learn the uncertainty and noise model, we build a differentiable IMU integrator and a differentiable covariance propagator on manifolds. Unlike the previous data-driven covariance prediction models [84, 85] that only predict the covariance of the output translation within a fixed window, we predict the accumulated covariance for a long-time pre-integration. Moreover, we jointly train the uncertainty module with the noise correction module, which can capture better IMU features and benefit both tasks.

In the experiments, we demonstrate the adaptability to a full spectrum of IMUs encompassing various grades and modalities, as illustrated in Figure 4.2. To validate the effectiveness of our learned covariance in sensor fusion, we design a pose graph optimization (PGO) system that fuses the AirIMU outputs with the GPS signals. To the best of our knowledge, we are the first to train a deep neural network that models IMU sensor uncertainty through differentiable covariance propagation. In summary, the main contributions of this paper are:

- This paper proposes AirIMU, a hybrid method that leverages the data-driven method to model the non-deterministic error, and the kinematic motion model to ensure generalizability in novel environments.
- We propose to jointly train the IMU noise and uncertainty through the differentiable in-

tegrator and covariance propagator. This training strategy improves 17.8% of the IMU preintegration results in Subt-MRS benchmark [64].

- Our method demonstrates adaptability across multiple grades of IMUs, from low-cost automotive-grade IMUs to high-end navigation-grade IMUs. We further evaluate AirIMU on the large-scale visual terrain navigation dataset ATLO [88] cruising over 262 km. Additionally, we illustrate the effectiveness of the learned uncertainty in a GPS-IMU PGO experiment, achieving a 31.6% improvement in optimization.
- We develop a tool package for the differentiable IMU integrator and covariance propagator that supports batched operations and the cumulative product, which speeds up the training process. This package is grounded on *PyPose* [60] library and the tutorial is now open-source at: <https://pypose.org/tutorials/imu>

4.2 Related Works

4.2.1 Traditional Inertial Odometry

Traditional model-based IO algorithms utilize the IMU kinematic motion model [70] to estimate the relative motion and are often integrated with other auxiliary sensors like GPS, Camera [3], and LiDAR [89] with filter-based method or batched optimization. To deploy IO on mobile robots, especially those equipped with low-cost IMUs, modeling IMU deterministic errors is indispensable [72], which generally include the additive sensor bias, scale factors, and axis misalignment [90].

To address the non-deterministic error, factors like nonlinearity, g-sensitivity [80], size effect [91], vibration [76], thermal effect [92], and Coriolis force [93] are rigorously addressed. These methods, however, rely on expensive motion simulators such as shakers and rate tables [94], which is not available in practice. To calibrate the low-cost IMUs, the IO algorithms utilize the IMU measurements under static conditions [72, 78], or auxiliary sensors like the camera [73, 95] and LiDAR [96] for calibration. Kalibr [73] is one of the most widely used calibration algorithms, calibrating the IMU intrinsic and extrinsic parameters between the camera and the IMU. However, all these calibration methods focus on modeling the linear behavior of the IMU sensor under static conditions or controlled environments, which often struggle to fully capture the noise inherent in the natural environment and sensor defects. This reveals a significant gap in the current IMU calibration and the practices of IO.

4.2.2 Learning Inertial Odometry

A growing trend in leveraging a data-driven model for IO has been witnessed, such as IMU odometry network [71, 97], EKF-fused IMU odometry [84, 85, 98] and learning-based IMU calibration [99]. In RIDI [97], a supervised Support Vector Machine (SVM) classifier is introduced to determine IMU placement modes (e.g., handheld, bag) using accelerometer and gyroscope measurements. Mode-specific Support Vector Regressions (SVRs) are then employed to estimate pedestrian velocities based on motion types, enabling corrective adjustments to sensor readings. IONet [71] introduced deep learning to regress the translation within a fixed temporal window. Building on this concept, RoNIN [81] improves the accuracy of the data-driven IO with multiple

neural network architectures, including Temporal Convolutional Networks (TCN) and Long Short-Term Memory (LSTM). RIO [98] explores rotation-equivalence as a self-supervisory mechanism for training IO models. These studies utilize data-driven motion estimation models to replace the IMU integration process, thereby reducing the drift associated with IMU double integration. Inspired by these IMU odometry network studies, TLIO [84] integrates the data-driven motion estimation model with the IMU integrator through a stochastic-cloning Extended Kalman Filter (EKF). This integrated approach optimizes position, orientation, and sensor biases, solely with IMU data. IDOL [85] further extends this framework with a data-driven orientation estimator, generating more accurate results with better consistency. Although these methods are promising in pedestrian datasets like RoNIN [81], the motion estimation heavily relies on prior knowledge about human movements, such as the consistent walking pattern of pedestrians and constant velocity assumption, which fail to generalize to other platforms like drones and vehicles. Moreover, these studies often assume the IMU measurement is transformed to the world coordinates or aligning the y-axis with gravity, which is impractical in real scenarios.

Inspired by the advancements of the data-driven methods in modeling unknown parameters, prior researchers employ neural networks to implicitly calibrate the IMUs [87, 99, 100]. Brossard et al. [99] train a Convolutional Neural Network (CNN) to denoise the low-frequency error in gyroscopes. Zhang et al. [87] employed a Long Short-Term Memory (LSTM) network to preprocess the raw IMU, and subsequently channeled the preprocessed IMU measurements into a filter-based visual-inertial odometry system. Deep IMU Bias [83], utilizes neural networks to model the sensor bias, and incorporates the learned bias as an additional constraint in the bundle adjustment. These studies showcase the potential of data-driven methods in modeling IMU errors, especially when the sensor’s physical conditions are difficult to capture. However, none of the referenced algorithms have explicitly tackled the challenge of modeling the uncertainty for IMU integration. When deploying these data-driven models, the predefined covariance parameters can no longer fit the learned IMU sensor model, undermining the IO’s accuracy and robustness.

4.2.3 IMU uncertainty model

The uncertainty model in IO, which quantifies the reliability of inertial measurements, is the key to optimizing both filter performance and the batched optimization results. Traditional IO follows the kinematics model formulated in the IMU integration to propagate the covariance matrix from the predefined IMU uncertainty parameters [70]. A commonly used solution to calibrate the uncertainty parameters is the Allen deviation [101], which quantifies the sensor bias and bias instability by analyzing the variance of the sensor errors in a static position. However, these parameters often require empirical tuning and manual re-calibration during testing due to unknown environmental disturbances and mechanical imperfections.

Recent researches in data-driven methods demonstrate the potential to model the uncertainty model with neural network [56]. The AI-IMU [82] proposes a dead-reckoning algorithm for vehicles, which assumes that the lateral and vertical velocities are approximately null. To model the confidence degree of these null assumptions, the authors train a CNN network from IMU to predict two constant uncertainty parameters corresponding to lateral and vertical velocity. DualProNet [102] is another attempt to use a deep neural network to predict the uncertainty parameters in IMUs. This work, however, requires high-frequency ground truth covariance labels for all IMU measurements, which are generally non-available during practice. In TLIO [84], the authors jointly train a covariance model of the predicted velocity with a fixed temporal window,

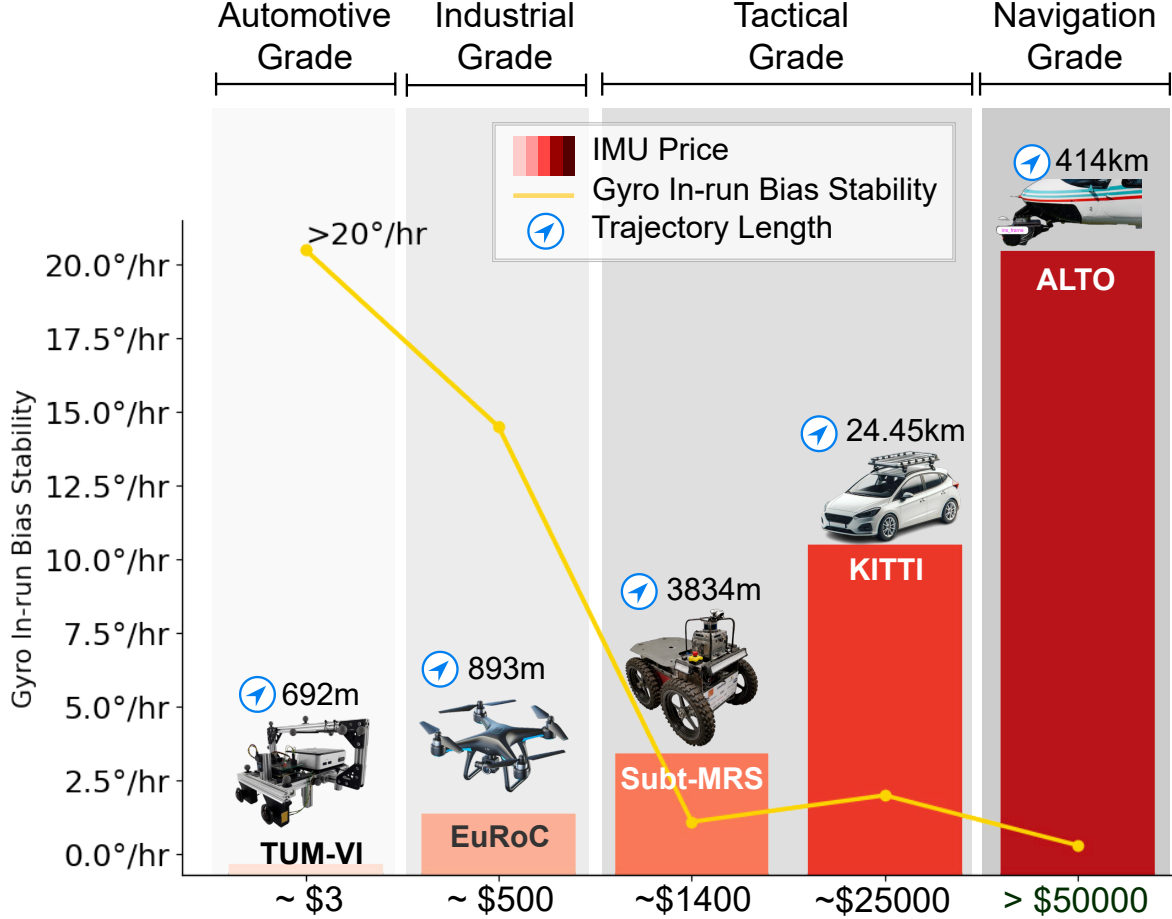


Figure 4.2: We evaluate our algorithm on IMUs with different prices and drift rates. Sensor quality is evaluated based on the gyroscope’s in-run bias stability metric. Our evaluation spans the full spectrum of IMUs, including automotive-grade, industrial-grade, tactical-grade, and navigation-grade IMUs.

which is later fused in an EKF.

In response to these challenges, our method leverages the differentiable covariance propagation to learn the high data-rate uncertainty. This approach alleviates the need for high-frequency ground truth labels and bridges the gap between uncertainty modeling and practical limitations.

4.3 AirIMU Model

To balance the accuracy and generalizability, the AirIMU model decouples the non-deterministic error model from the precisely defined kinematic motion model, as depicted in Figure 4.3. Specifically, it utilizes a data-driven model to capture the error and uncertainty inherent in the IMU, while employing the kinematic-based method to integrate the state and covariance. By supervising the integrated state and covariance, the AirIMU jointly trains the sensor error and uncertainty models throughout the differentiable integrator. To accelerate the training speed with long sequences, we propose a tree-like structure for cumulative operation and redesign the covariance propagation to facilitate batched operation and parallel computing. The refined state and covariance outputs are subsequently incorporated into the PGO module, where they are fused with

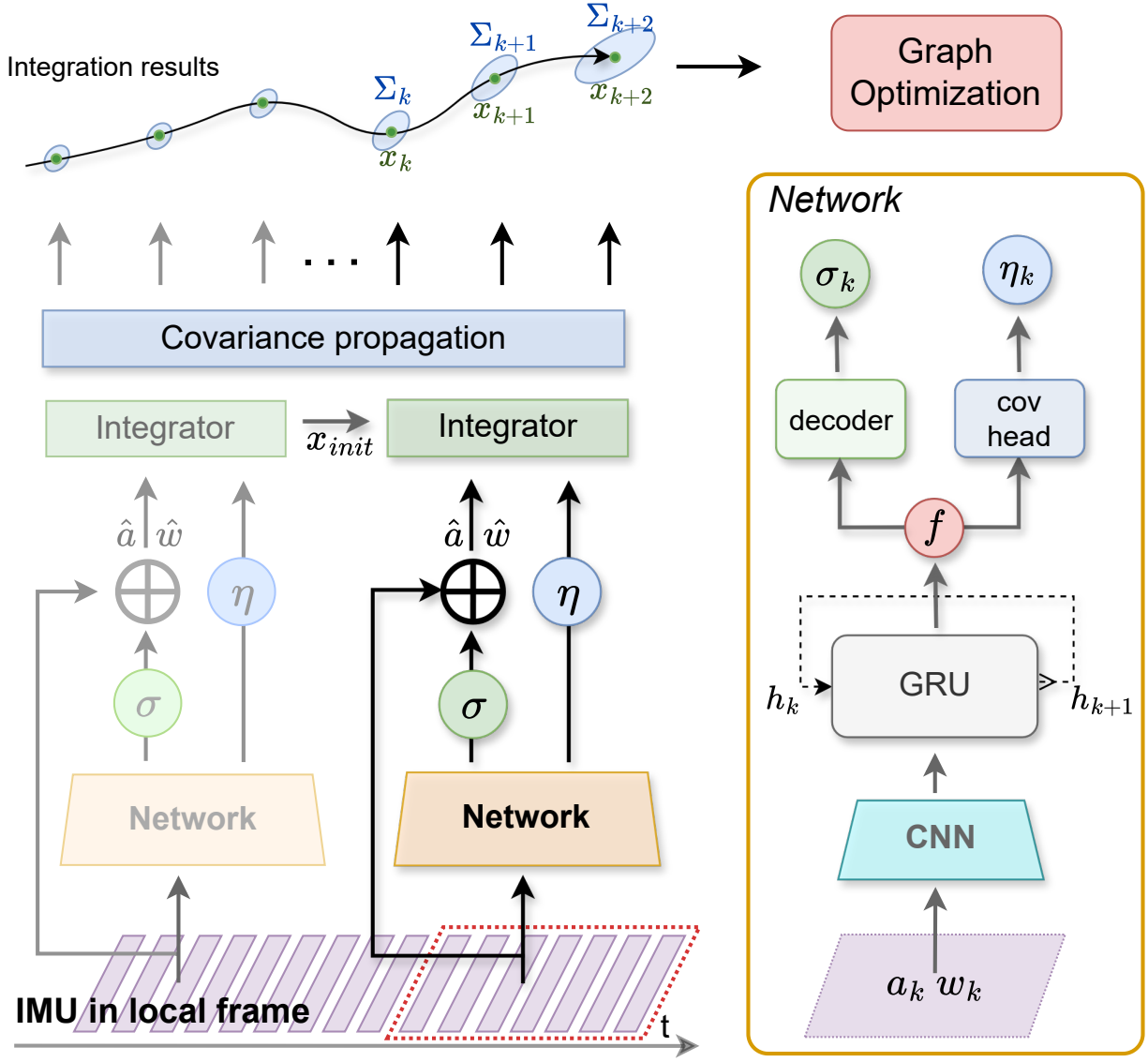


Figure 4.3: Right: We employ a CNN-GRU encoder to capture IMU features (f) from raw IMU accelerometer (a_k) and gyroscope (w_k) measurements. Subsequently, these features are decoded to obtain IMU corrections ($\sigma_{acc}, \sigma_{gyro}$). **Left:** We add the raw measurements with learned corrections, generating corrected IMU measurements ($\hat{a}_k$ and $\hat{w}_k$). The learned uncertainties (η) are propagated to estimate the corresponding covariance matrix (Σ).

auxiliary sensors such as GPS.

Overall, we show the data-driven model in Sec. 4.3.1, the kinematic model in Sec. 4.3.2, loss functions in Sec. 4.3.3. Moreover, we detail our efforts of accelerating the integration and covariance propagation in Sec. 4.3.4. Lastly, we present the PGO module that fuses the AirIMU with auxiliary sensors like GPS in Sec. 4.3.5.

4.3.1 Data-driven module

As shown in the right part of Figure 4.3, the AirIMU model utilizes an encoder-decoder network structure to process raw IMU data, which includes three-axis acceleration a and three-axis angular velocity w , both measured in the body frame B . The network outputs corrections for acceleration and angular velocity, denoted as ${}^B\sigma^{acc}$ and ${}^B\sigma^{gyro}$, respectively. Additionally, it calculates the corresponding uncertainty measures for the gyroscope η^{gyro} and accelerometer η^{acc} .

To leverage and understand the hidden relationship between the IMU noise and uncertainty, we train a shared encoder network to capture the common representations of the raw IMU measurements for both tasks. In the encoder model, we first utilize a 1D CNN network to learn the lower-level features like the pattern of the accelerometer and gyroscope. To capture the temporal dependency from the previous IMU data, we combine the CNN network with gated recurrent unit (GRU) layers to generate the IMU feature.

With the shared features, we build a correction decoder to estimate the error ${}^B\sigma^{gyro}, {}^B\sigma^{acc}$ in the following:

$$[{}^B\sigma^{gyro}, {}^B\sigma^{acc}] = f_{\pi}({}^Bw, {}^Ba), \quad (4.1)$$

where π represents the learnable parameters of the decoder. For the covariance, we train another decoder with a similar structure to predict the uncertainty of each corresponding frame. This uncertainty is predicted by a covariance model denoted as:

$$[\eta^{gyro}, \eta^{acc}] = \Sigma_{\theta}({}^Bw, {}^Ba), \quad (4.2)$$

where θ represents the learnable parameters of the covariance decoder model.

Overall, the corrected acceleration and angular velocity ${}^B\tilde{a}$ and ${}^B\tilde{w}$, are formulated as the summation of the original input ${}^Ba, {}^Bw$, the network corrections ${}^B\sigma^{acc}, {}^B\sigma^{gyro}$, and noise of the measurements:

$${}^B\tilde{a} = {}^Ba + {}^B\sigma^{acc} + \eta_k^{acc}, \quad {}^B\tilde{w} = {}^Bw + {}^B\sigma^{gyro} + \eta_k^{gyro}. \quad (4.3)$$

Combined with these outputs with the learned uncertainty $\eta_k^{gyro}, \eta_k^{acc}$, we utilize the IMU kinematic model to estimate the states Wx_j , and more importantly, to propagate the corresponding covariance $\Sigma_{i,j}$.

4.3.2 Differentiable Integration and Covariance Module

Labeling the ground truth of the high-frequency data requires expensive motion simulators, which is infeasible due to the cost. Therefore, AirIMU builds a differentiable integrator and covariance propagator to supervise the data-driven module throughout the integration and covariance propagation.

The IMU pre-integration follows Newton's kinematic laws, which are well-defined in [70]. The integrated state x_k includes orientation R , velocity v , and position p within the reference frame

$\{W\}$ in the k -th frame. The integration from i -th frame to j -th frame is formulated as:

$$\begin{aligned}\Delta R_{ij} &= \int_{t \in [t_i, t_j]} {}^B \tilde{w}_k dt \\ \Delta v_{ij} &= \int_{t \in [t_i, t_j]} R_k^B \tilde{a}_k dt \\ \Delta p_{ij} &= \int \int_{t \in [t_i, t_j]} R_k^B \tilde{a}_k dt^2 ,\end{aligned}\tag{4.4}$$

where the ΔR_{ij} , Δv_{ij} and Δp_{ij} denotes the increments of rotation, velocity and position from frame i to frame j respectively. Given these increments, the j -th state of the robot ${}^W x_j$ in the world frame $\{W\}$ can be predicted through:

$$\begin{aligned}{}^W R_j &= \Delta R_{ij} \otimes {}^W R_i \\ {}^W v_j &= {}^W R_i * \Delta v_{ij} + {}^W v_i \\ {}^W p_j &= {}^W R_i * \Delta p_{ij} + {}^W p_i + {}^W v_i \Delta t_{ij},\end{aligned}\tag{4.5}$$

where the ${}^W R_i$, ${}^W v_i$, and ${}^W p_i$ are the state of i -th frame.

The propagation of the IMU covariance also follows the IMUs' kinematic model. Following the conventions established by Forster et al. [70], the covariance of the predicted state is characterized as: $[\delta \phi_k^T, \delta v_k^T, \delta p_k^T]^T \sim \mathcal{N}(0_{9 \times 1}, \Sigma_k)$, where the $\delta \phi_k$, δv_k and δp_k represents the zero-mean Gaussian noise for rotation, velocity and position. We apply a similar covariance propagation strategy as [70] to model the covariance matrix $\Sigma_{i,i}$ of the IMU integration from frame i to frame $i+1$ through iterative propagation. Starting from the initial covariance matrix $\Sigma_{i,i} = 0_{9 \times 9}$, we predict the covariance matrix as:

$$\Sigma_{i,i+1} = A \Sigma_{i,i} A^T + B_g \text{diag}(\eta_i^{gyro}) B_g^T + B_a \text{diag}(\eta_i^{acc}) B_a^T, \tag{4.6}$$

where state transition matrix A is defined as:

$$A = \begin{bmatrix} \Delta R_{kk+1}^T & 0_{3 \times 3} & 0_{3 \times 3} \\ -\Delta R_{ik}({}^B \hat{a}_k^{\wedge}) \Delta t & I_{3 \times 3} & 0_{3 \times 3} \\ -1/2 \Delta R_{ik}({}^B \hat{a}_k^{\wedge}) \Delta t^2 & I_{3 \times 3} \Delta t & I_{3 \times 3} \end{bmatrix}. \tag{4.7}$$

The B matrix addresses the frame-by-frame uncertainty of the acceleration and the angular velocity.

$$B_g = \begin{bmatrix} J_r^k \Delta t \\ 0_{3 \times 3} \\ 0_{3 \times 3} \end{bmatrix}, \quad B_a = \begin{bmatrix} 0_{3 \times 3} \\ \Delta R_{ik} \Delta t \\ 1/2 \Delta R_{ik} \Delta t^2 \end{bmatrix}. \tag{4.8}$$

The covariance matrix is 9×9 matrix, and we denote it as:

$$\Sigma_{i,j+1} = \begin{bmatrix} \Sigma_{i,j+1}^r & 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & \Sigma_{i,j+1}^v & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} & \Sigma_{i,j+1}^p \end{bmatrix}, \tag{4.9}$$

where $\Sigma_{i,j+1}^r$, $\Sigma_{i,j+1}^v$ and $\Sigma_{i,j+1}^p$ are 3×3 matrix representing the covariance of rotation, velocity,

and position respectively.

4.3.3 Loss Function and Training strategy

To supervise the learnable model in AirIMU, we design the state loss (4.10) for noise correction and covariance loss (4.11) for uncertainty estimation. Given the differentiable integrator defined in (4.4) and (4.5), we train the denoising module $f_\pi(w, a)$ by supervising the state loss with the position, orientation, and translation in the world frame $\{W\}$.

$$\begin{aligned} L_r &= \|\log({}^W R_i \otimes {}^W \hat{R}_i)\|_h, \\ L_v &= \|{}^W v_i - {}^W \hat{v}_i\|_h, \\ L_p &= \|{}^W p_i - {}^W \hat{p}_i\|_h. \end{aligned} \quad (4.10)$$

The $\|\cdot\|_h$ denotes the Huber function, chosen for its robustness in penalizing the large error terms, particularly in lengthy integration sequences. We transform the orientation error to log space in the loss function (4.10) to supervise the rotation in the manifold, which reduces the single value during training.

To supervise the covariance, we use a similar loss function derived in Russell et al. [56]. For the covariance of rotation $\Sigma_{i,j}^r$, velocity $\Sigma_{i,j}^v$ and position $\Sigma_{i,j}^p$, we design the covariance losses as:

$$\begin{aligned} L_r^{cov} &= \frac{1}{2} \left(\left\| \log({}^W R_j \otimes {}^W \hat{R}_j) \right\|_{\Sigma_{i,j}^r}^2 + \ln(\det \Sigma_{i,j}^r) \right), \\ L_v^{cov} &= \frac{1}{2} \left(\|{}^W v_j - {}^W \hat{v}_j\|_{\Sigma_{i,j}^v}^2 + \ln(\det \Sigma_{i,j}^v) \right), \\ L_p^{cov} &= \frac{1}{2} \left(\|{}^W p_j - {}^W \hat{p}_j\|_{\Sigma_{i,j}^p}^2 + \ln(\det \Sigma_{i,j}^p) \right), \end{aligned} \quad (4.11)$$

where y is the ground-truth label and $f(w, a)$ output the integrated state. The $\Sigma_\theta(x)$ denotes the covariance model that predicts the covariance matrix given the learned uncertainty η . To simplify and stabilize the training process, our training focuses on the diagonal components of the covariance matrix. The final loss function can be denoted as:

$$L = L_r + L_v + L_p + \varepsilon(L_r^{cov} + L_v^{cov} + L_p^{cov}), \quad (4.12)$$

where we set the weight of the covariance loss as $\varepsilon = 1 \times 10^{-3}$. We chose this parameter to mitigate the impact of relatively small covariance values, which otherwise lead to large covariance losses.

In contrast to previous methods that utilized data augmentation techniques such as random bias and rotation equivalence, we avoid applying data augmentation to sensor model training. Our approach preserves the original noise in the raw data, which is critical for the AirIMU model to learn the noise pattern in the real sensor. For most datasets, we set the length of the training segment as 1000 frames (5 seconds). This duration allows us to capture both the drift and the long-term features essential for IMU integration.

For the optimizer, we employ the Adam with a learning rate of 1×10^{-3} with a weight decay 1×10^{-4} . The weighting of each loss term and learning rate may be adjusted to accommodate the specific characteristics of different types of IMU, particularly with different bias instability of

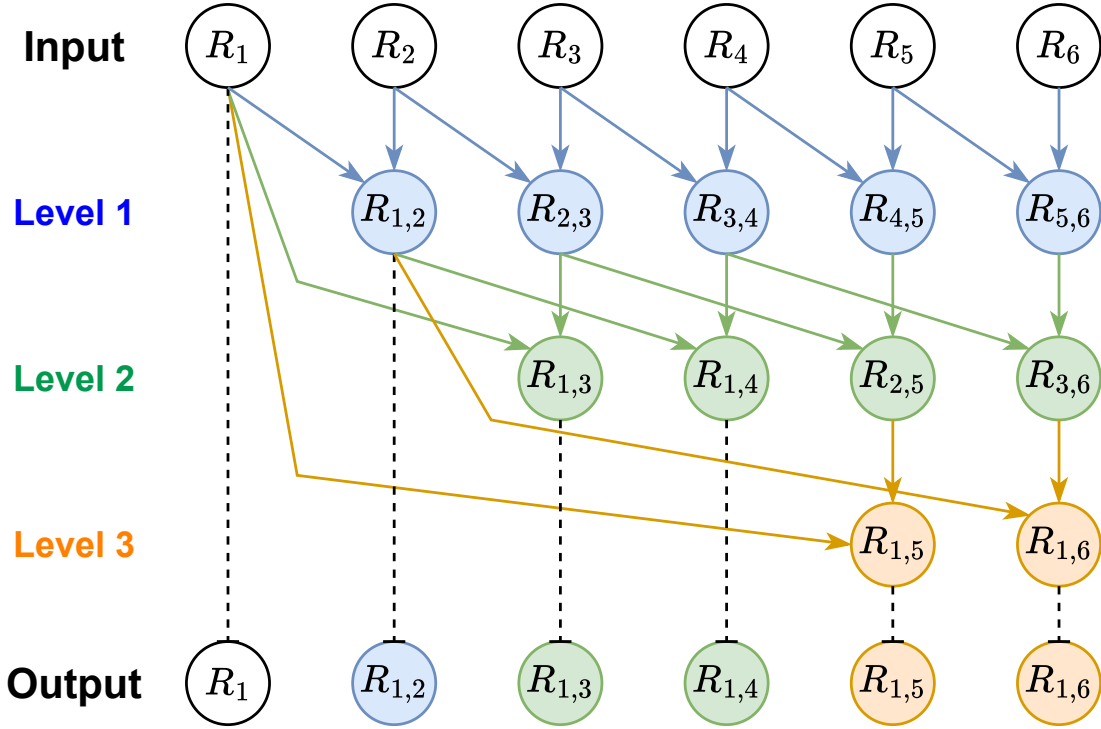


Figure 4.4: This graph presents the parallel scan we use to accelerate the cumulative product in our IMU integration. Given a sequence of gyroscope input from R_1 to R_n , the algorithm iterates through $\log(n)$ layers. In each layer i , it updates the last $N - 2^i$ elements by multiplying the first 2^i elements. In this algorithm, each layer is a parallel batch that supports parallel computation.

gyroscopes and accelerometers.

4.3.4 Implementation & Acceleration

In our training process, we extend the supervision period to enhance the effectiveness of the training, which is particularly crucial with high-end IMUs that only exhibit drift after long periods of integration. While extending the training over long segments can be time-consuming, our AirIMU model addresses this challenge by fully leveraging parallel computing capabilities powered by *PyTorch*. In this section, we introduce the algorithm we designed to accelerate the cumulative product and cumulative summation to facilitate parallel computation during training. Additionally, we reformulate the iterative covariance propagation to enable batched operation for long segments. These efforts not only accelerate the training process but also optimize the efficiency of real-time applications.

Specifically, the IMU preintegration and covariance propagation defined in (4.4) and (4.6) relies heavily on the cumulative product of $SO3$ on manifolds and cumulative summation of the acceleration. To accelerate the integration, we design a parallel scan algorithms as shown in Figure 4.4, where each node represents an element in $SO3$. In this implementation, the integration with a length of N can be separated into $\log_2(N)$ iterations. During each iteration i , it updates

each node with the preceding node positioned 2^i place before it, which can be parallel computed. This approach achieves $O(\log N)$ time complexity while maintaining the $O(N)$ memory usage.

Different from the segment tree strategy proposed by Brossard et al. [99], which predicts only the final state $R_{1,6}$ of the gyroscope integration, our approach is designed to calculate every intermediate result throughout the integration within the same level of time. The intermediate results significantly accelerates the calculation of the matrix of A , and thereby accelerates the covariance propagation. Moreover, we not only use this approach in gyroscope integration, but also for accelerometer integration and the covariance propagation. Overall, we integrate this algorithm into two operators: the `cumsum` for the cumulative summation and the `cumprod` for the cumulative product of the $SO3$ on manifolds.

The conventional solution outlined in (4.6) iteratively calculates the covariance of each state. This iterative approach, while methodically sound, requires performing covariance propagation thousands of times, even for supervising brief segments lasting a few seconds. Such an iterative method is inefficient for training, leading to significant computational burdens. To address this inefficiency and fully utilize the parallel computing capabilities, we reformulate the covariance propagation for $\Sigma_{i,j}$ to two steps.

First of all, we calculate the matrix list $C_{i,j}^A$ with $\prod_{k=j}^i A_k$ through the `cumprod` of the matrix A_k , and the matrix list $C_{i,j}^B$ consisting the initial covariance $\Sigma_{i,i}$ and the matrix B .

$$\begin{aligned} C_{i,j}^A &= [\prod_{k=j}^i A_k, \prod_{k=j}^{i+1} A_k, \dots, A_j, I_{9 \times 9}], \\ C_{i,j}^B &= [\Sigma_{i,i}, B_i, \dots, B_{j-1}, B_j]. \end{aligned} \quad (4.13)$$

Secondly, the covariance $\Sigma_{i,j+1}$ can be calculated by the product over $C_{i,j}^A$ and $C_{i,j}^B$:

$$\Sigma_{i,j+1} = C_{i,j}^A \text{diag}(C_{i,j}^B) C_{i,j}^{A^T}. \quad (4.14)$$

With this re-implementation, we are now able to supervise the IMU integration over extended sequences. For instance, in the helicopter navigation scenario described in Sec. 4.4.5, our model is trained with unusually long segments that exceed one minute in duration. This extended supervision period captures the drift inherent in the navigation-grade IMU after long-term integration, which is crucial for achieving reliable navigation in aerial environments.

4.3.5 IMU-GPS PGO system

To recover the inevitable drift from inertial odometry, auxiliary sensors like GPS are employed to fuse with the IMU measurements in the PGO. When fusing the IMU measurements with other sensor, the covariance of the IMU integration $\Sigma_\theta(x)$, plays a critical role in the fusion. In this section, we describe the construction of a PGO that incorporates both the AirIMU model and GPS. We demonstrate how leveraging the learned covariance from AirIMU enhances the optimization process. This integrated system not only compensates for the drift inherent in IMU pre-integration but also effectively utilizes the complementary sensors, resulting in more reliable and accurate navigation results.

In Figure 4.5, we show the system pipeline and the factor graph incorporating GPS signals.

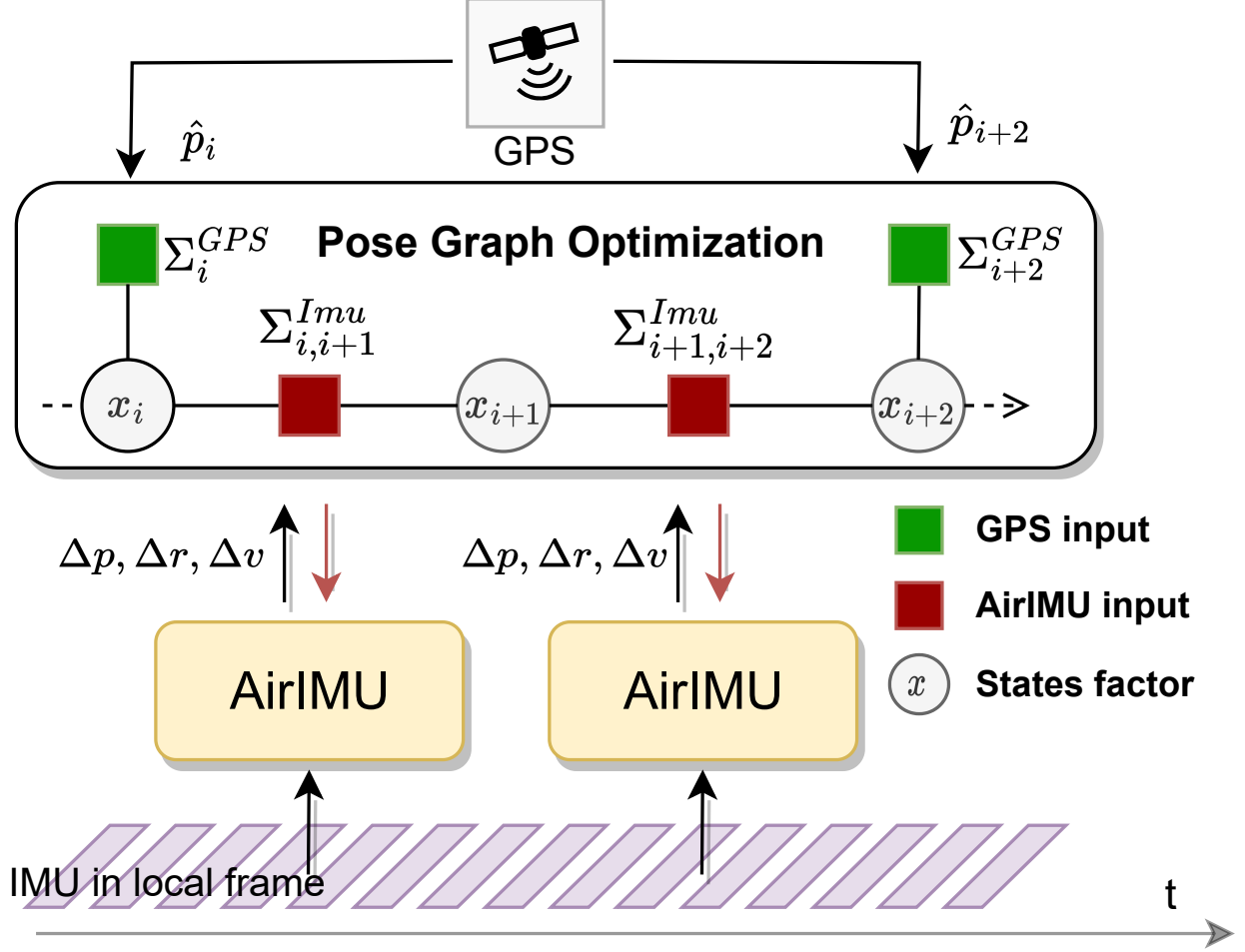


Figure 4.5: Pose graph optimization of a GPS signal with IMU pre-integration. The state of the graph is defined as x_i constrained by two different observations: IMU (Δp , Δv and Δr) and GPS ($\hat{p}_i$). We back-propagate the gradient through the IMU pre-integration and covariance propagation to learn the integration and noise model.

The objective of the PGO is to optimize the state of the robot x , which is modeled as the factor in our PGO. From the AirIMU, we take the pre-integrated increments (Δp , Δv , and Δr) as the observation. These observations are then propagated alongside the initial state $x_i(R_i, v_i, p_i)$ to calculate the state $x_j(R_j, v_j, p_j)$, forming the basis to compute the IMU residual:

$$e_{ij}^{IMU}(\Delta v_{ij}, \Delta p_{ij}, \Delta r_{ij}) = \begin{bmatrix} \Delta R_{ij}^W R_i ({}^W R_j)^{-1} \\ {}^W v_i + {}^W R_i * \Delta v_{ij} - {}^W v_j \\ {}^W p_i + {}^W R_i * \Delta p_{ij} - {}^W p_j \end{bmatrix}. \quad (4.15)$$

To calculate the corresponding covariance matrix Σ_{ij}^{IMU} of the IMU integration. We leverage the predicted uncertainty from the network $[\eta_k^{acc}, \eta_k^{gyro}] = \Sigma_\theta(w, a)$ and employ it in (4.6) to propagate the covariance matrix Σ_{ij}^{IMU} . Since the initial state x_i is updated after each optimization, we also update the corresponding covariance $\Sigma_{i,i}$ of the i^{th} state during the optimization. This ensures the covariance of the state is bounded, giving the other sensors' measurements and

Table 4.1: Dataset summary showing the comprehensive evaluation across different IMU grades and platforms

Datasets	Duration	IMU	Modality
EuRoC [65]	22m29s	ADIS16448	Drone
TUM-VI [103]	13m31s	BMI160	Handheld
SubtMRS [64]	2h52m	Epson M-G365	Ground robot
KITTI [66]	43m44s	OXTS RT 3000	Vehicle
ALTO [89]	2h12m	NG LCI-1	Helicopter

advancing the reliability.

We define the GPS signal as $\hat{p}_i \in R^3$ to constrain the state of position ${}^W p_i$, where the residual of GPS measurements are:

$$e_i^{GPS}({}^W \hat{p}_i) = {}^W \hat{p}_i - {}^W p_i \quad (4.16)$$

where the measurement of the GPS is ${}^W p_i$ with a corresponding covariance matrix defined as a diagonal matrix Σ_i^{GPS} . The optimization with both the $e_i^{GPS}(\hat{p}_i)$ and $e_{ij}^{IMU}(\Delta v_{ij}, \Delta p_{ij}, \Delta r_{ij})$ can be formulated as:

$$X^* = \arg \min_X \left\{ \sum_{k \in G} \|e_k^{GPS}\|_{\Sigma_k^{GPS}}^2 + \sum_{l \in L} \|e_l^{IMU}\|_{\Sigma_l^{IMU}}^2 \right\} \quad (4.17)$$

where G and L is the set of GPS and IMU observations and the X is the factor. We can solve this optimization by Levenberg-Marquardt algorithms.

4.4 Experiments

As shown in Figure 4.2, we comprehensively evaluate our method across a full spectrum of IMUs, from low-cost automotive-grade to high-end navigation-grade IMUs, each featured on different platforms and modalities. The price and gyro in-run bias stability are plotted, revealing a clear correlation between cost and performance stability. At the high end of the spectrum, we have the NG LITEF LCI-1 in the ALTO dataset [88] with a cost exceeding \$50,000, showcasing exceptional bias stability in helicopters. The tactical-grade IMUs, like the OXTS RT3003 in the KITTI dataset [66] and the Epson M-G365 in the Subt-MRS [64], are priced at approximately \$25,000 and \$1,400, deployed on UGVs and vehicles respectively. On the low-cost end, we feature the industrial-grade IMU BMI 160 on drones at around \$500 and the automotive-grade IMU ADIS 16448 at \$3 used in handheld platforms, demonstrating the feasibility of using low-cost IMUs for certain applications. These datasets evaluate the adaptability of our model not only across multiple grades of IMUs but also across various platforms and modalities.

In our experiments, we demonstrate the dual purposes of the AirIMU: state estimation and uncertainty estimation.

State Estimation: We demonstrate the AirIMU model’s superior performance in IMU integration, outperforming both learning-based IMU de-noising algorithms and traditional IMU calibration algorithms. Notably, the AirIMU model achieves a tenfold increase in accuracy compared to learning-based inertial odometry methods in short-term integration. We also demonstrate the potential of the AirIMU model in GPS-denied helicopter navigation using high-end IMUs. This assessment spans a cruising distance over 262 km, providing a realistic and challenging environment for evaluating our method.

Learned Uncertainty: We assess our model with a particular emphasis on the potency of the learned uncertainty. An ablation study focused on a GPS-PGO experiment reveals an average improvement of 31.6% in utilizing the learned uncertainty. Furthermore, we conduct an ablation study to highlight the benefits of jointly training the uncertainty module with the IMU noise correction. This dual training approach significantly improves the network’s ability to extract features from IMUs, resulting in a performance enhancement of 17.8% compared to models solely trained on noise correction.

4.4.1 Evaluation Metrics

We define the following metrics to evaluate the performance of our model over assigned time intervals:

Rotational Root Mean Squared Error (R-RMSE, rad),

$$\text{R-RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\| \log \left(\hat{R}_{i,i+\Delta t}^T R_{i,i+\Delta t} \right) \right\|_2^2}, \quad (4.18)$$

calculates the RMSE of the rotations over the assigned time intervals on the manifold space.

Positional Root Mean Squared Error (P-RMSE, m)

$$\text{P-RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\| p_{i+\Delta t} - p_i - R_i \hat{R}_i^T (\hat{p}_{i+\Delta t} - \hat{p}_i) \right\|_2^2}, \quad (4.19)$$

calculates the RMSE of the relative displacements over the assigned time intervals.

4.4.2 IMU Integration Performance

Figure 4.6 shows comprehensive integration results across all evaluated datasets. AirIMU consistently outperforms traditional calibration methods and learning-based approaches across different IMU grades and integration periods. The method shows particular strength in:

- **Cross-IMU Adaptability:** Superior performance across automotive-grade to navigation-grade IMUs
- **Long-term Integration:** Reduced drift accumulation over extended periods
- **Real-world Robustness:** Effective handling of environmental disturbances and non-deterministic errors

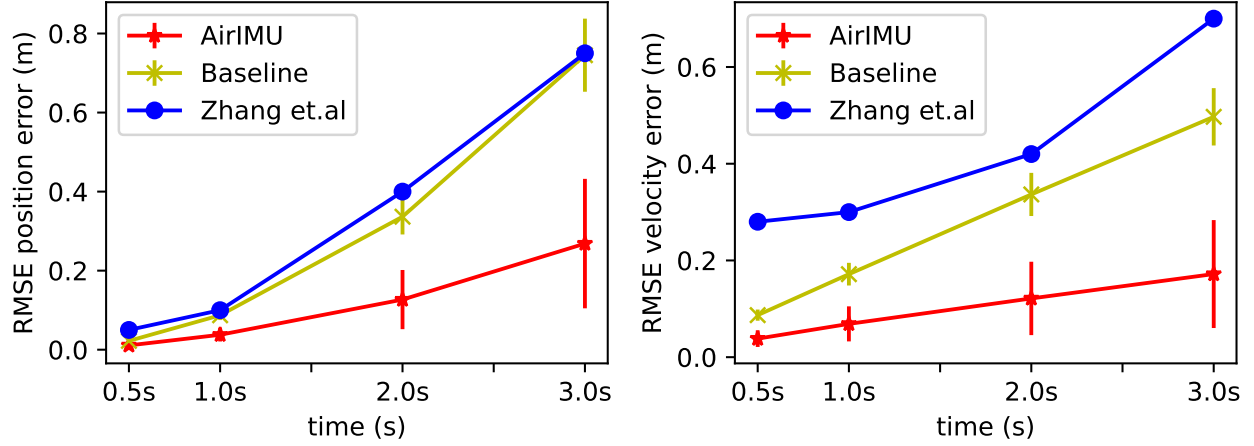


Figure 4.6: Integration results comparison showing AirIMU performance across different integration periods and datasets. AirIMU consistently outperforms baseline methods across various IMU grades and integration durations.

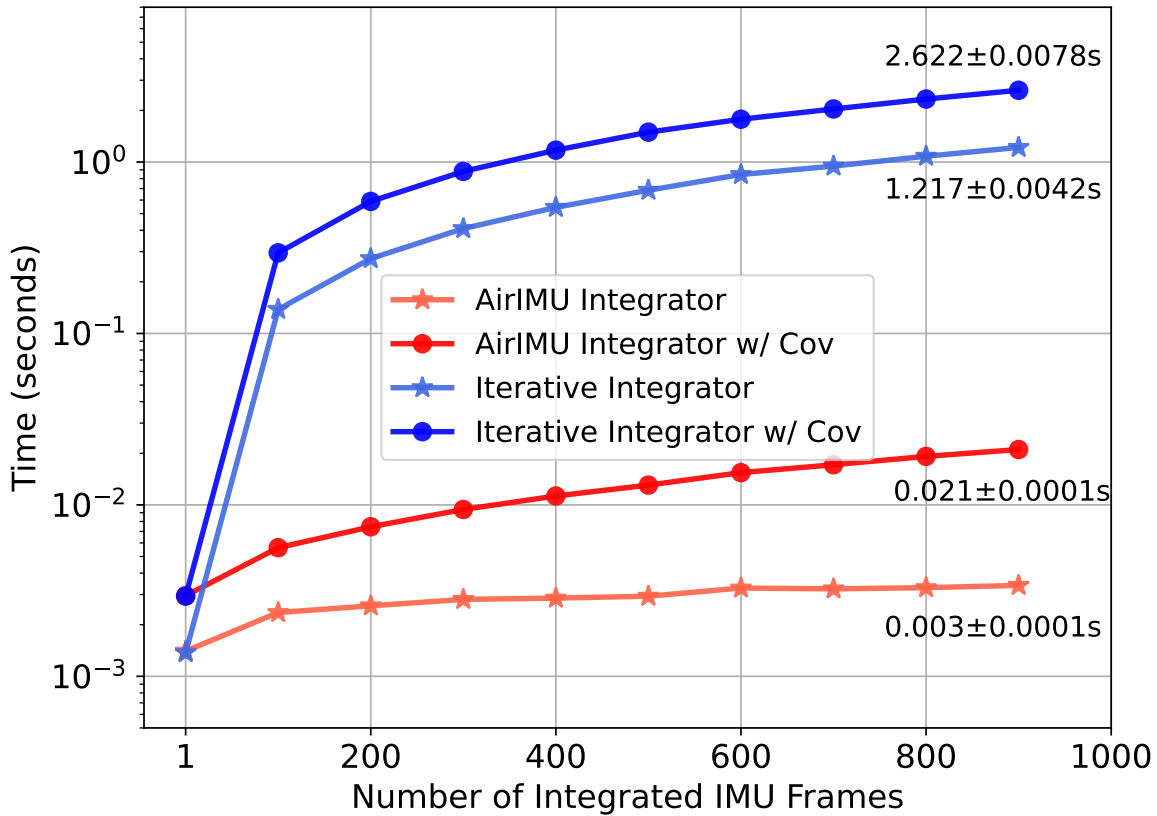


Figure 4.7: Time analysis of the AirIMU integrator compared to iterative methods. We integrate IMU input with different lengths (from 1 frame to 1000 frames) and display the time consumption on a log scale. The AirIMU integrator demonstrates a hundredfold improvement in efficiency.

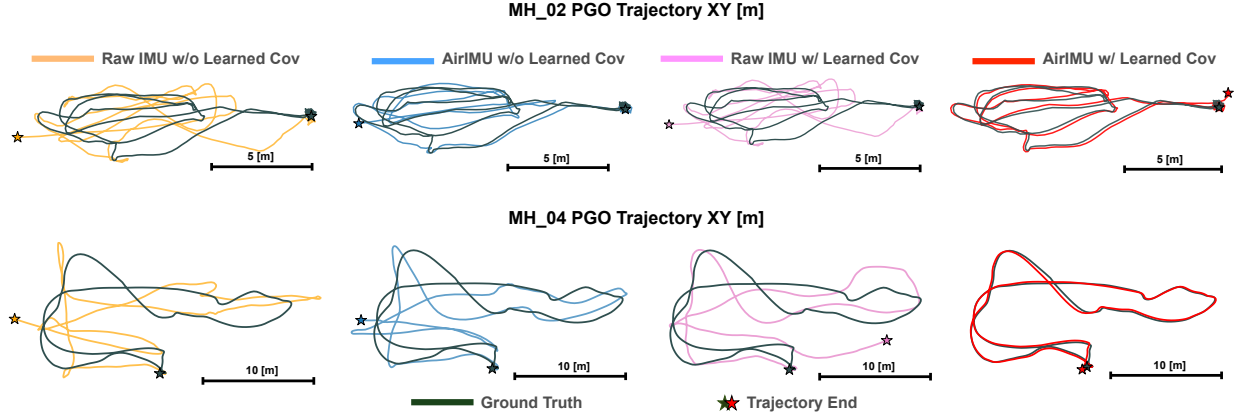


Figure 4.8: PGO results visualization showing the effectiveness of learned uncertainty in GPS-IMU fusion. The learned covariance from AirIMU enables more accurate and robust optimization compared to fixed uncertainty models.

4.4.3 Computational Efficiency

Figure 4.7 demonstrates the computational advantages of our parallel implementation. The differentiable integrator with parallel scan algorithms achieves:

- **100x speedup** compared to iterative covariance propagation methods
- **$O(\log N)$ complexity** for cumulative operations through tree-structured algorithms
- **Batched processing** enabling efficient training on long sequences

4.4.4 GPS-IMU Pose Graph Optimization

The integration with GPS through pose graph optimization (Figure 4.8) validates the effectiveness of learned uncertainty models. Key results include:

- **31.6% improvement** in GPS-IMU fusion accuracy using learned covariance
- **Adaptive uncertainty:** Dynamic adjustment to sensor conditions and environmental factors
- **Robust optimization:** Better convergence and stability in challenging scenarios

4.4.5 Large-Scale Navigation Evaluation

The helicopter navigation evaluation (Figure 4.9) on the ALTO dataset demonstrates AirIMU’s capability for large-scale, long-duration navigation:

- **262 km trajectory:** Extended navigation without GPS dependency
- **Navigation-grade IMU:** Effective utilization of high-end sensor capabilities
- **Real-world conditions:** Robust performance under aerial navigation challenges

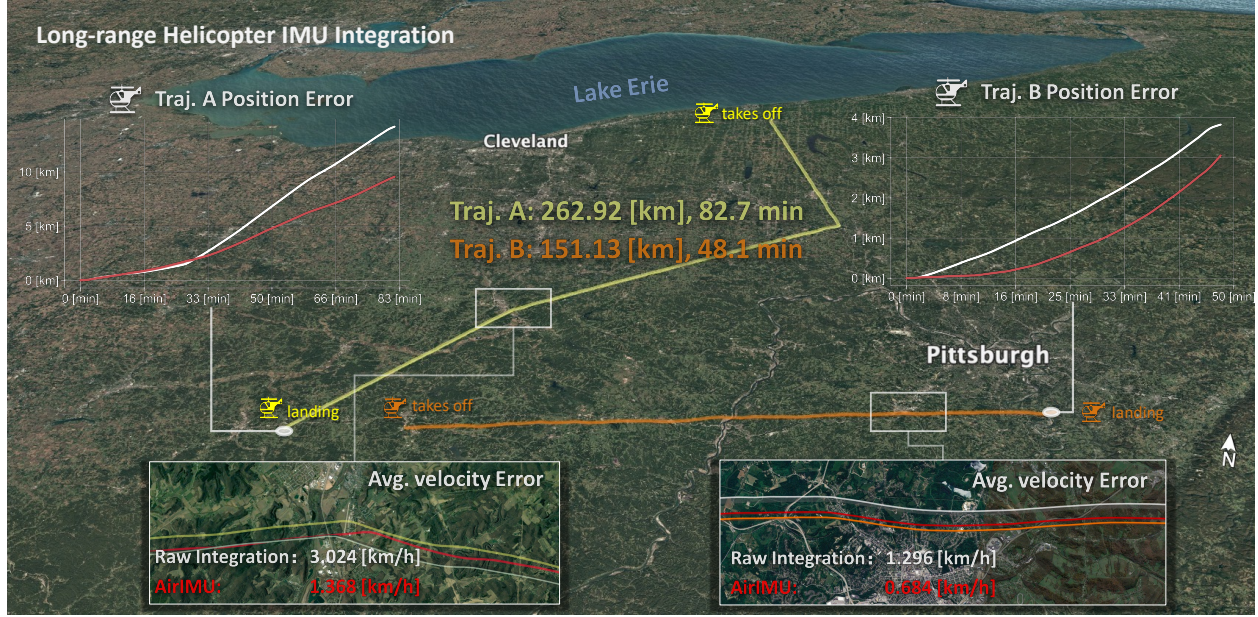


Figure 4.9: Near-earth helicopter navigation trajectory visualization over 262 km using navigation-grade IMUs. AirIMU enables precise long-term navigation in GPS-denied environments.

4.5 Limitation and Discussion

We propose a hybrid method that can adapt to a full spectrum of IMUs. However, we’ve observed that the AirIMU model trained on a single IMU does not readily generalize to a different IMU with different specifications and operating frequencies. For instance, a model trained for the tactical-grade IMU **Epson G365**, used in the SubT-MRS dataset, does not apply to the automotive-grade IMU **ADIS16448** in the TUMVI dataset due to their different physical design. Our current model is specialized, focusing on a single IMU’s unique noise profile and uncertainty characteristics rather than attempting to create a one-size-fits-all model. If we need to deploy the AirIMU model to a new type of IMU, fine-tuning with new IMU data is necessary.

In the future, we may explore the potential for generalizing through different sensors within the same models or types of IMUs. For example, whether the sensor model trained on one **Epson G365** can be adapted to other **Epson G365** IMUs, potentially a more versatile application to scale up in manufacturing the same types of IMU.

4.6 Conclusion

We propose AirIMU, a hybrid IO system with a differentiable integrator and covariance propagator. To demonstrate the efficacy of the learned covariance model, we conduct an ablation study on a GPS PGO experiment, revealing that the learned covariance decreased the loss of PGO by 31%. We evaluate our model in all spectrums of IMUs’ grades, spanning the entry-level IMUs (Automotive Grade) to the high-end IMUs (Navigation Grade). Our experiments demonstrate that the AirIMU model achieves outstanding performance across the board. The evaluation also includes IMUs operating in a wide range of environments and featuring multiple agent modalities. We find that AirIMU not only excels at regular indoor environments but

also performs well with a ground vehicle dataset over 2 km and a large-scale helicopter dataset over 414 km. Additionally, we open-source the differentiable IMU integrator and the covariance propagation, which is implemented in the *PyPose* framework. Given these advancements, we are optimistic that the proposed method can provide a strong foundation for further research in inertial navigation systems and multi-sensor fusion.

Chapter 5

AirIO: Learning Inertial Odometry with Enhanced Feature Observability

5.1 Introduction

This chapter presents AirIO (AI-enhanced Inertial Odometry), which explores enhanced IMU feature observability for robust motion estimation. Building on the uncertainty learning established in AirIMU, AirIO demonstrates how learning-based approaches enable uncertainty-aware inertial odometry in GPS-denied environments.

5.2 Enhanced IMU Feature Observability

5.2.1 Problem Formulation

Traditional inertial odometry suffers from limited observability of certain motion states, particularly in scenarios with constrained motion patterns. This section addresses how enhanced feature extraction can improve observability.

[Content to be developed based on AirIO research]

5.2.2 Learning-Based Feature Enhancement

[Content to be developed]

5.3 Uncertainty-Aware Inertial Odometry

5.3.1 Integration with Visual Perception

Building on the uncertainty models from MAC-VO and AirIMU, this section demonstrates how uncertainty-aware inertial odometry can be integrated with visual perception systems.

[Content to be developed]

5.3.2 Real-Time Uncertainty Estimation

[Content to be developed]

5.4 Experimental Validation

5.4.1 GPS-Denied Navigation

[Content to be developed]

5.4.2 Cross-Platform Evaluation

[Content to be developed]

5.5 Conclusion

[Content to be developed based on research outcomes]

Part II

Part II: Uncertainty-Aware Multi-Modal Fusion and Cross-Platform Adaptation

This part extends single-sensor uncertainty learning to multi-modal fusion scenarios. Building on the individual sensor models from Part I, we develop cross-modal uncertainty correlations (MACVIO) and demonstrate cross-platform adaptation capabilities (MACIO). These chapters establish uncertainty as the theoretical foundation for optimal sensor fusion and enable deployment across diverse robotic platforms.

Chapter 6

MACVIO: Learning-Based Visual-Inertial SLAM with Cross-Modal Uncertainty Fusion

6.1 Introduction

This chapter presents MACVIO (Metrics-Aware Cross-Modal Visual-Inertial Odometry), which extends the single-sensor uncertainty learning from MAC-VO and AirIMU to multi-sensor fusion scenarios. MACVIO demonstrates how cross-modal uncertainty correlations enable adaptive fusion weights and uncertainty-driven sensor selection.

6.2 Cross-Modal Uncertainty Correlation Learning

6.2.1 Problem Formulation

Traditional visual-inertial SLAM systems rely on fixed fusion weights or simple heuristics for sensor selection. This approach fails to capture the dynamic correlation between visual and inertial uncertainty estimates that varies across environments and motion patterns.

[Content to be developed based on MACVIO research]

6.2.2 Learning Cross-Modal Dependencies

Building on the uncertainty models established in MAC-VO (Chapter 3) and AirIMU (Chapter 4), this section develops methods for learning the correlation structure between visual and inertial uncertainties.

[Content to be developed]

6.3 Adaptive Sensor Fusion Weights

6.3.1 Uncertainty-Driven Weight Adaptation

[Content to be developed]

6.3.2 Real-Time Fusion Strategy

[Content to be developed]

6.4 Uncertainty-Driven Sensor Selection

6.4.1 Selective Integration Framework

[Content to be developed]

6.4.2 Performance under Sensor Degradation

[Content to be developed]

6.5 Visual-Inertial SLAM with Learned Uncertainty

6.5.1 System Architecture

[Content to be developed]

6.5.2 Loop Closure with Uncertainty Estimates

[Content to be developed]

6.6 Experimental Validation

6.6.1 Benchmark Datasets

[Content to be developed]

6.6.2 Comparison with State-of-the-Art

[Content to be developed]

6.6.3 Cross-Platform Evaluation

[Content to be developed]

6.7 Conclusion

[Content to be developed based on research outcomes]

Chapter 7

MACVIO: Learning-Based Visual-Inertial SLAM with Cross-Modal Uncertainty Fusion

7.1 Introduction

This chapter presents MACVIO (Metrics-Aware Cross-Modal Visual-Inertial Odometry), which extends the single-sensor uncertainty learning from MAC-VO and AirIMU to multi-sensor fusion scenarios. MACVIO demonstrates how cross-modal uncertainty correlations enable adaptive fusion weights and uncertainty-driven sensor selection.

7.2 Cross-Modal Uncertainty Correlation Learning

7.2.1 Problem Formulation

Traditional visual-inertial SLAM systems rely on fixed fusion weights or simple heuristics for sensor selection. This approach fails to capture the dynamic correlation between visual and inertial uncertainty estimates that varies across environments and motion patterns.

[Content to be developed based on MACVIO research]

7.2.2 Learning Cross-Modal Dependencies

Building on the uncertainty models established in MAC-VO (Chapter 3) and AirIMU (Chapter 4), this section develops methods for learning the correlation structure between visual and inertial uncertainties.

[Content to be developed]

7.3 Adaptive Sensor Fusion Weights

7.3.1 Uncertainty-Driven Weight Adaptation

[Content to be developed]

7.3.2 Real-Time Fusion Strategy

[Content to be developed]

7.4 Uncertainty-Driven Sensor Selection

7.4.1 Selective Integration Framework

[Content to be developed]

7.4.2 Performance under Sensor Degradation

[Content to be developed]

7.5 Visual-Inertial SLAM with Learned Uncertainty

7.5.1 System Architecture

[Content to be developed]

7.5.2 Loop Closure with Uncertainty Estimates

[Content to be developed]

7.6 Experimental Validation

7.6.1 Benchmark Datasets

[Content to be developed]

7.6.2 Comparison with State-of-the-Art

[Content to be developed]

7.6.3 Cross-Platform Evaluation

[Content to be developed]

7.7 Conclusion

[Content to be developed based on research outcomes]

Part III

Part III: Uncertainty-Aware SLAM

*This part focuses on **uncertainty-aware SLAM**. Building on learned odometry and IMU uncertainty, we develop multi-sensor SLAM formulations that explicitly model measurement covariance, cross-factor correlations, and calibration drift. We investigate how uncertainty calibration impacts data association, loop closure, and robust optimization, and how these elements scale across environments and platforms.*

Part IV

Part IV: Uncertainty-Driven Adaptation, Self-Supervision, and Representation Learning

This part proposes adaptation- and representation-focused methods that leverage uncertainty to improve inertial perception and its integration with broader robot learning systems. We target self-supervised IO, contact-aware legged odometry, and transformer-based IMU representations that interface with vision foundation models and reinforcement learning.

Proposed Work A — MAC-IO: Uncertainty-Guided Self-Supervised Inertial Odometry

- Learn motion-adaptive IMU noise/bias models; use calibrated uncertainty as self-supervision for IO without ground-truth trajectories.
- RQ-A1: How can predictive uncertainty calibrate pseudo-labels and losses to stabilize self-supervised IO across devices and motions?
- RQ-A2: How should bias/scale drift be modeled and updated online under uncertainty-aware objectives?

Proposed Work B — Contact-Aware Legged Odometry via Differentiable EKF (Humanoids)

- Learn a contact/foot–terrain model and integrate it into a differentiable EKF for IMU odometry under gait transitions and slip.
- RQ-B1: How to jointly learn contact states and their uncertainty so the EKF can down-weight unreliable phases (e.g., slip, toe-off)?
- RQ-B2: Does end-to-end differentiation through the EKF improve observability and reduce drift in contact-rich locomotion?

Proposed Work C — IMU Tokenizer: Transformer-Based Inertial Representation for VFM and RL

- Tokenize raw IMU streams to learned embeddings; align with visual tokens for sensor fusion and downstream policy learning.
- RQ-C1: What tokenization (patching, events, frequency bands) best preserves dynamics while enabling cross-modal alignment with vision?
- RQ-C2: Can uncertainty-aware IMU tokens improve sample efficiency and robustness in RL and visuomotor policies?

Chapter 8

MACIO: Cross-Platform Adaptation with Self-Supervised Uncertainty Models

8.1 Introduction

This chapter presents MACIO (Multi-Agent Cross-Platform Intelligent Odometry), which addresses the challenge of deploying learned uncertainty models across different robotic platforms. MACIO develops self-supervised domain transfer methods that enable zero-shot deployment with automatic sensor configuration.

8.2 Self-Supervised Domain Transfer

8.2.1 Problem Formulation

Current perception systems fail to generalize across robotic platforms due to variations in sensor configurations, mounting positions, and calibration parameters. This section formulates the cross-platform adaptation problem for uncertainty models.

[Content to be developed based on MACIO research]

8.2.2 Domain Adaptation Framework

Building on the uncertainty models from MACVIO (Chapter 6), this section develops methods for transferring learned uncertainty correlations across different platform configurations.

[Content to be developed]

8.3 Platform-Agnostic Uncertainty Models

8.3.1 Invariant Feature Learning

[Content to be developed]

8.3.2 Cross-Modal Calibration

[Content to be developed]

8.4 Automatic Sensor Configuration

8.4.1 Self-Calibrating Systems

Traditional SLAM systems require extensive manual calibration for each new platform. This section presents methods for automatic sensor parameter discovery and configuration.

[Content to be developed]

8.4.2 Online Adaptation Mechanisms

[Content to be developed]

8.5 Zero-Shot Deployment on New Platforms

8.5.1 Transfer Learning Strategy

[Content to be developed]

8.5.2 Performance Validation Framework

[Content to be developed]

8.6 Experimental Validation

8.6.1 Cross-Platform Benchmarking

[Content to be developed]

8.6.2 Real-World Deployment Studies

Validation across different robotic platforms including:

- Quadrotor UAVs with different sensor suites
- Ground vehicles with varying camera configurations
- Handheld devices with consumer-grade sensors
- Legged robots with dynamic sensor mounting

[Content to be developed]

8.6.3 Comparison with Manual Tuning

[Content to be developed]

8.7 Conclusion

[Content to be developed based on research outcomes]

Chapter 9

Contact Learning from Legged Odometry

9.1 Introduction

This chapter presents contact learning from legged odometry, which explores how perception uncertainty propagation is critical for contact-rich robotics. We demonstrate force-vision fusion for legged locomotion with contact-aware perception models that bridge the gap between uncertain perception and physical interaction.

9.2 Uncertainty Propagation from Perception to Action

9.2.1 Problem Formulation

Traditional robotic systems treat perception and action as separate modules, failing to account for how perception uncertainty affects contact-rich behaviors. This section formulates the uncertainty propagation problem from perception systems to physical interaction.

[Content to be developed based on contact learning research]

9.2.2 Theoretical Framework

Building on the uncertainty models from previous chapters (MAC-VO, AirIMU, MACVIO, MACIO), this section develops the theoretical foundation for propagating perceptual uncertainty through contact dynamics.

[Content to be developed]

9.3 Contact-Rich State Estimation with Uncertain Perception

9.3.1 Legged Odometry Framework

Legged robots provide a unique opportunity to learn contact dynamics while simultaneously performing state estimation. This section presents methods for contact-aware state estimation under perception uncertainty.

[Content to be developed]

9.3.2 Uncertainty-Contact Interaction Models

[Content to be developed]

9.4 Force-Vision Fusion for Legged Locomotion

9.4.1 Multi-Modal Contact Detection

Combining visual perception uncertainty with tactile feedback for robust contact detection and characterization.

[Content to be developed]

9.4.2 Contact-Aware Visual SLAM

Traditional visual SLAM assumes static environments, but legged locomotion involves dynamic contact with the environment. This section presents contact-aware visual SLAM that leverages leg odometry.

[Content to be developed]

9.5 Learning Contact-Aware Perception Models

9.5.1 Self-Supervised Learning from Locomotion

[Content to be developed]

9.5.2 Cross-Modal Contact Learning

Learning the relationship between visual appearance, contact forces, and locomotion outcomes for improved perception.

[Content to be developed]

9.6 Experimental Validation

9.6.1 Legged Robot Platforms

Validation on different legged robot platforms including:

- Quadrupedal robots (e.g., ANYmal, Spot)
- Bipedal robots for human-like locomotion
- Multi-legged robots for complex terrains

[Content to be developed]

9.6.2 Terrain Characterization

[Content to be developed]

9.6.3 Comparison with Traditional Approaches

[Content to be developed]

9.7 Applications to Manipulation

9.7.1 Contact-Rich Manipulation Tasks

Extending contact learning from locomotion to manipulation scenarios where perception uncertainty affects contact quality.

[Content to be developed]

9.7.2 Dual-Arm Coordination

[Content to be developed]

9.8 Conclusion

[Content to be developed based on research outcomes]

Chapter 10

Uncertainty-Aware Perceptive Locomotion

10.1 Introduction

This chapter presents uncertainty-aware perceptive locomotion, which explores how uncertain perception requires fundamentally different locomotion strategies. We develop uncertainty-driven planning that learns from physical interaction feedback to enable safe exploration and generalizable skills.

10.2 Uncertainty-Driven Locomotion Planning

10.2.1 Problem Formulation

Traditional locomotion planning assumes perfect perception, which fails in real-world scenarios where perception uncertainty is inherent. This section formulates locomotion planning under perception uncertainty.

[Content to be developed based on research progress]

10.2.2 Uncertainty Propagation in Motion Planning

[Content to be developed]

10.3 Learning from Physical Interaction Feedback

10.3.1 Contact-Aware Perception Models

[Content to be developed]

10.3.2 Safe Exploration Strategies

[Content to be developed]

10.4 Generalizable Locomotion Skills

10.4.1 Cross-Platform Skill Transfer

[Content to be developed]

10.4.2 Adaptation to New Environments

[Content to be developed]

10.5 Experimental Results

10.5.1 Simulation Studies

[Content to be developed]

10.5.2 Hardware Validation

[Content to be developed]

10.6 Conclusion

[Content to be developed based on research outcomes]

Chapter 11

Conclusion and Future Directions

11.1 Summary of Contributions

This dissertation has presented a comprehensive framework for learning-based uncertainty estimation in robotic perception systems. The key contributions spanning individual sensor modalities, multi-sensor fusion, and physical interaction include:

- **Visual Perception Uncertainty (MAC-VO):** Metrics-aware covariance learning for stereo visual odometry that generalizes across diverse visual conditions
- **Inertial Perception Uncertainty (AirIMU):** Data-driven IMU preintegration uncertainty models that adapt to platform-specific noise characteristics
- **Inertial Odometry Enhancement (AirIO):** Learning-based approaches for uncertainty-aware inertial odometry with enhanced feature observability
- **Multi-Modal Fusion (MACVIO):** Cross-modal uncertainty correlation learning enabling adaptive sensor fusion and uncertainty-driven sensor selection
- **Cross-Platform Adaptation (MACIO):** Self-supervised domain transfer methods for uncertainty models enabling zero-shot deployment
- **Contact-Rich Robotics:** Learning from legged odometry for contact-aware perception and force-vision fusion
- **Uncertainty-Aware Locomotion:** Uncertainty-driven planning strategies that learn from physical interaction feedback

11.2 Thesis Impact and Validation

The research presented in this thesis has demonstrated significant impact in the robotics community:

11.2.1 Publication Record

- MAC-VO received the **Best Paper Award at ICRA 2025**, validating the importance of learned uncertainty in visual perception
- AirIMU contributions to inertial uncertainty modeling have been recognized across multiple venues
- The framework has enabled cross-platform deployment without manual parameter tuning

11.2.2 Technical Achievements

The unified theme of learning-based uncertainty estimation has enabled:

- Robust generalization across diverse robotic platforms and environmental conditions
- Elimination of manual parameter tuning in SLAM systems
- Advancement toward fully automated spatial intelligence for autonomous systems

11.3 Addressing the Research Questions

This thesis has systematically addressed the seven research questions posed in the introduction:

RQ1-RQ3 (Individual Sensor Modalities): Demonstrated that learned uncertainty models significantly outperform hand-tuned parameters across visual, inertial, and odometry applications.

RQ4-RQ5 (Multi-Sensor Systems): Established uncertainty as the theoretical foundation for optimal sensor fusion and enabled cross-platform adaptation through self-supervised learning.

RQ6-RQ7 (Contact-Rich Robotics): Showed that uncertainty propagation is critical for contact-rich scenarios and that uncertainty-aware planning enables safer and more generalizable robotic behaviors.

11.4 Future Directions

The research presented opens several exciting avenues for future exploration:

11.4.1 Foundation Model Integration

- Leveraging large-scale pre-trained models for uncertainty estimation
- Zero-shot uncertainty calibration using foundation model priors
- Multimodal uncertainty learning across vision, language, and action spaces

11.4.2 Scaling to Complex Environments

- Urban-scale deployment of uncertainty-aware SLAM systems
- Integration with semantic understanding and scene graph representation
- Long-term autonomy with continual uncertainty model adaptation

11.4.3 Theoretical Foundations

- Formal guarantees for learned uncertainty estimates
- Connections between uncertainty learning and optimal control theory
- Information-theoretic frameworks for uncertainty-driven sensor selection

11.5 Toward Fully Automated Spatial Intelligence

The ultimate vision driving this research—making SLAM engineering obsolete through full automation—remains an ambitious but achievable goal. The contributions presented here represent significant steps toward:

- **Parameter-Free Systems:** Elimination of manual tuning through learned uncertainty models
- **Platform Agnosticism:** Automatic adaptation to new sensor configurations and robotic platforms
- **Environmental Robustness:** Generalization across diverse operational conditions without human intervention
- **Intelligent Decision Making:** Uncertainty-driven planning and control that adapts to perception reliability

11.6 Final Remarks

Learning uncertainty for spatial intelligence represents a paradigm shift from traditional hand-engineered approaches to data-driven, adaptive systems. The research presented in this thesis demonstrates that uncertainty is not merely a byproduct of perception but rather a fundamental signal that enables robust, generalizable, and intelligent robotic behavior.

As robotic systems become increasingly deployed in unstructured, real-world environments, the ability to understand and reason about perceptual uncertainty becomes critical for safe and reliable operation. The framework developed here provides the theoretical foundations and practical tools necessary for this next generation of spatially intelligent systems.

The journey toward fully automated spatial intelligence continues, but the path forward is now clearly illuminated by the principles of learning-based uncertainty estimation established in this work.

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